

GaN-based Temperature Sensing and Over-temperature Protection ICs

By

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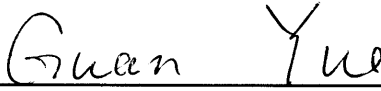
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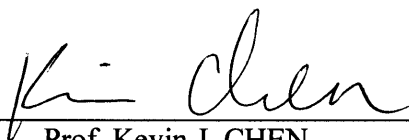
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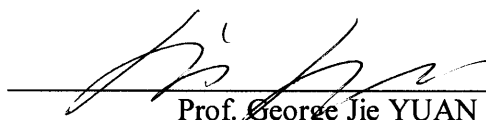
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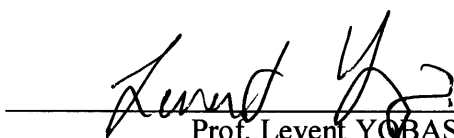
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Index Table

GaN-based Temperature Sensing and Over-temperature Protection ICs.....	i
Authorization	ii
GaN-based Temperature Sensing and Over-temperature Protection ICs.....	iii
Acknowledgements	iv
Index Table	vi
List of Figures	viii
List of Tables	x
GaN-based Temperature Sensing and Over-temperature Protection ICs.....	1
Abstract.....	1
Chapter 1 Introduction.....	3
1.1 Background of GaN-based Technology	3
1.2 Introduction of Temperature Sensor	6
1.2.1 Resistance Temperature Detector (RTDs)	7
1.2.2 Thermistor	8
1.2.3 Thermocouples	9
1.2.4 Sensor ICs on Silicon-based Technology.....	11
1.3 GaN-based Platform.....	14
1.3.1 AlGaN/GaN HEMT	14
1.3.2 AlGaN/GaN SBD	15
1.3.3 AlGaN/GaN L-FER	16
1.4 Motivation of This Work.....	17
1.5 Objective of This Work.....	19
Chapter 2 Temperature-dependent Characteristic of GaN-based Devices	21
2.1 Motivation.....	21
2.2 Fabrication	21
2.3 Temperature-dependent Characterization.....	23
2.3.1 D-mode HEMT.....	23
2.3.2 E-mode HEMT	24
2.3.4 Schottky Barrier Diode	25
2.3.4 Lateral-field Effect Rectifier	26
2.4 Summary.....	27
Chapter 3 GaN-based temperature sensor	28
3.1 Motivation.....	28
3.2 Design of GaN-based temperature sensor	28

3.3	Fabrication	31
3.4	Measurement.....	32
3.4.1	Temperature Dependent DC Measurement	32
3.4.2	Power Supply Rejection Ratio (PSRR).....	35
3.5	Summary.....	38
Chapter 4 GaN-based PTAT Temperature Sensor.....		39
4.1	Motivation for Design of PTAT Temperature Sensor	39
4.2	PTAT Temperature Sensor Structure and Operating Principle	42
4.2.1	All-D-mode Current Mirror.....	43
4.2.2	Pair of SBDs PTAT Temperature Sensor	44
4.3	Fabrication	46
4.4	Measurements and Result Analysis	47
4.4.1	Temperature Dependent DC Measurements	47
4.4.2	Power Supply Rejection Ratio (PSRR).....	51
4.5	Summary.....	53
Chapter 5 Over-temperature Protection.....		55
5.1	Motivation.....	55
5.2	Design and Working Principle of Over-temperature Protection.....	55
5.3	Fabrication	58
5.4	Measurement and Analysis.....	59
5.5	Summary.....	62
Chapter 6 Conclusion and Suggestion for Future Work		64
6.1	Conclusion	64
6.2	Suggestions for Future Work.....	65
REFERENCES.....		67

List of Figures

Fig. 1.1.1 Application of GaN-based devices.....	4
Fig. 1.2.1 Platinum RTD resistance vs. temperature (straight line for reference) [3]..	8
Fig. 1.2.2 Thermistor resistance vs. temperature:	9
(a) linear scale. (b) logarithmic scale [3].	9
Fig. 1.2.3 Schematic of thermocouple.....	10
Fig. 1.2.4 Diagram of a silicon-based integrated smart temperature sensor [2].	12
Fig. 1.2.5 Schematic of temperature sensor IC on silicon-based technology.....	13
Fig. 1.3.1 Cross-section of D-mode HEMT and E-mode HEMT	15
Fig. 1.3.2 Cross-section of SBD	16
Fig. 1.3.3 Cross-section of L-FER	17
Fig 1.4.1. GaN-based power electronic platform.	18
Fig 1.4.2. A diagram of smart power functional blocks.....	18
Fig. 2.2.1 Process flow of GaN-based devices fabrication.....	23
Fig. 2.3.1 Temperature dependent I - V characteristic of D-mode HEMT.....	24
Fig. 2.3.2 Temperature dependent I - V characteristic of E-mode HEMT.....	25
Fig. 2.3.3 Temperature dependent I - V characteristic of SBD.....	26
Fig 2.3.5 Temperature dependent I - V characteristic of L-FER.	27
Fig. 3.2.1 Design of GaN-based temperature sensor.....	29
Fig. 3.2.2 Operating principle of GaN-based temperature sensor	30
Fig. 3.2.3 Temperature-dependent characteristics of (a) M1 (b) LFERs.	31
Fig. 3.3.1 SEM image of GaN-based temperature sensor.	32
Fig. 3.4.1 Temperature dependent V_o - V_{DD} relationship.....	33
Fig. 3.4.3 Schematic of on-chip testing platform.....	36
Fig. 3.4.4 PSSR of temperature sensor as a function of frequency at $V_{DD} = 8$ V	37

Fig. 3.4.5 Temperature-dependent PSRR of GaN-based temperature sensor	38
Fig. 4.1.1 Schematic of basic PTAT current generator.	39
Fig. 4.1.2 4T cell self-biased differential common gate amplifier based PTAT voltage generator.	41
Fig. 4.2.1 Structure of PTAT temperature sensor.	42
Fig. 4.2.2 Schematic of all-D-mode current mirror.....	43
Fig. 4.2.3 Sensing part of PTAT temperature sensor	44
Fig. 4.3.1 SEM image of PTAT temperature sensor	47
Fig. 4.4.1 Temperature dependent V_o - V_{DD} characteristic	48
Fig. 4.4.2 V_o -T characteristic of PTAT temperature sensor	49
Fig. 4.4.3 Temperature dependent I - V characteristic of sensing SBD	49
Fig. 4.4.4 $V_{x,y}$ - V_{DD} relation at RT (left) and 275 °C (right)	50
Fig. 4.4.5 Measured PSRR as V_{DD} =14 V at room temperature	52
Fig. 4.4.6 Temperature dependent PSRR characteristic at 10 kHz.....	53
Fig. 5.2.1 Functional block diagram of over-temperature protection	56
Fig. 5.2.2 Structure of over-temperature protection.....	57
Fig. 5.3.1 SEM imaging: (a) temperature sensor; (b) comparator	59
Fig. 5.4.1 Temperature dependent output voltage of over-temperature protection	61
(a) $V_{DD} = 10$ V, V_{REF} sweeps from 0 to 4 V; (b) $V_{DD} = 12$ V, V_{REF} sweeps from 0 to 4 V; (c) $V_{DD} = 10$ V, $V_{REF} = 2.07$ V; (d) $V_{DD} = 12$ V, $V_{REF} = 1.95$ V.....	61
Fig. 5.4.2 Output voltage with changeable critical temperature	62
(a) $V_{DD} = 10$ V; (b) $V_{DD} = 12$ V.....	62

List of Tables

Table 1-1 Properties of competing materials in semiconductor platforms.....	4
Table 1-2 Competitive advantages for commercial product of GaN devices.....	6
Table 1-3 Seebeck coefficient and temperature range for several thermocouple types	10
Table 3-1 The characteristics of V_o - T with different V_{DD}	35
Table 3-2 PSRR of GaN-based temperature sensor at room temperature.....	36
Table 3-3 Temperature-dependent PSRR of GaN-based temperature sensor	37
Table 4-1 PSRR of PTAT temperature sensor.....	51
Table 4-2 Temperature-dependent PSRR of PTAT temperature sensor	52
Table 4-3 Comparison of two types of sensors on GaN-based platform	53

GaN-based Temperature Sensing and Over-temperature Protection ICs

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Abstract

The wide bandgap gallium nitride (GaN) and related compound materials possess attractive properties including high breakdown electric field, high electron saturation velocity, relatively strong piezoelectric effect and excellent capability of operating at high temperatures. GaN-based electronic devices are quickly emerging as the game changer for many mission-critical applications including high-frequency power amplifiers, high-voltage power switches, high-temperature pressure and gas sensors, etc. For all these applications, it is essential to build temperature sensing and protection circuits using the GaN device technology, for the benefits of on-chip monitoring and protection.

In this work, the temperature-dependent characteristics of the GaN-based high electron mobility transistor (HEMT), Schottky barrier diode (SBD) and lateral-field effect rectifier (LFER) are investigated first. Based on the device characteristic, a GaN-based temperature sensor is designed to convert the temperature change into a voltage signal. The output voltage shows a stable negative temperature coefficient.

This sensing circuit exhibits a larger temperature sensing range than the silicon CMOS temperature sensors. Thus, an on-chip over-temperature protection circuit is demonstrated for the GaN smart power integrated circuits (IC). This protection circuit features monolithic integration of a temperature sensing circuit and a high-gain comparator circuit, and can be easily integrated with GaN power devices.

A proportional to absolute temperature (PTAT) sensor circuit is also designed, fabricated and tested using GaN technology for the first time. The output voltage, which is linearly proportional to temperature, shows a positive temperature coefficient. In this design, a HEMT-based current mirror circuit is proposed to substitute PMOS current mirror in the conventional CMOS PTAT sensor design, for compatibility with GaN-HEMT based technology. The Schottky barrier diode (SBD), readily available on the GaN HEMT platform, is used in place of the P-N junction in the CMOS technology.

The temperature sensing and protection circuits demonstrated in this work provide the essential analog functional blocks for the implementation of GaN power ICs and integrated sensors.

Chapter 1 Introduction

1.1 Background of GaN-based Technology

GaN, III-V compound semiconductor, and its alloy compounds have already been well developed implanted technologies in the LED optoelectronics industry in the blue to ultraviolet (UV) range of the spectrum. For about ten years, GaN-based devices also have been researched and applied in RF applications where they can replace silicon-based transistors for some specific applications especially in military applications such as radar, by the advantage of its high-frequency capability. Due to the superior material properties, such as strong piezoelectric effect and high-temperature tolerance, GaN have been investigated and demonstrated in sensing in high-temperature applications such as gas and pressure sensors in harsh environment. Recently, GaN also attracts lots of attentions in power electronics especially for high power applications by the advantages of high-power, high-breakdown field, and high efficiency. All these excellent and unique properties enable GaN semiconductors to be potential game changer for most of the key applications, especially for those applications where conventional compound and silicon semiconductors cannot be applied. Fig 1.1.1 shows a wide scope of applications where the GaN-based devices can be applied promisingly. Compared to silicon, GaN-based devices exhibit largely better figures for most of the key specifications, especially for high-power applications.

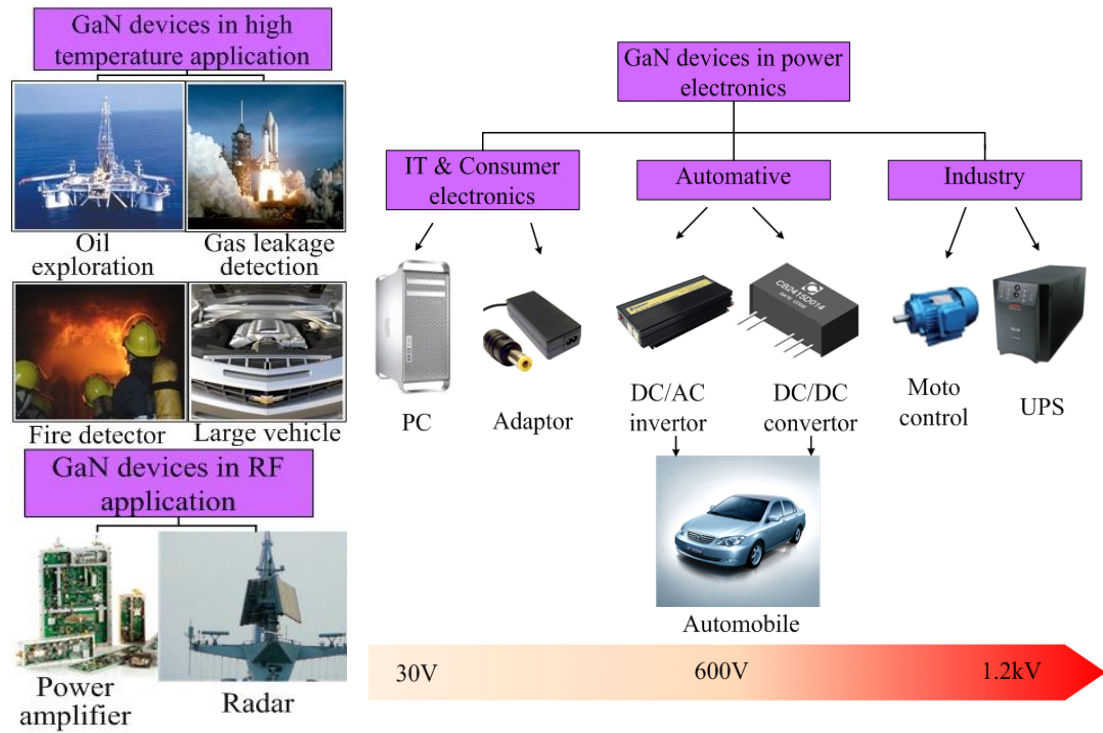


Fig. 1.1.1 Application of GaN-based devices.

For future high power applications, silicon power devices, after 30 years development, have almost reached its theoretical limits. Fortunately, GaN-based devices could substitute silicon-based devices to offer better performance in terms of: high operating temperature up to 700 °C, 10 times breakdown voltage, and 5 times switching frequency than silicon MOSFET. These superior performances can be directly guaranteed by 3 times bandgap, almost 2 times electron mobility and 2 times saturation electron velocity. The material properties of four widely used semiconductors are listed in Table 1-3 [1] - [4].

Table 1-1 Properties of competing materials in semiconductor platforms

Characteristics	Unit	Silicon	Gallium Arsenide (AlGaAs /InGaAs)	Silicon Carbide	Gallium Nitride (AlGaN /GaN)
Bandgap	eV	1.1	1.42	3.26	3.49

Electron Mobility at 300K	cm ² /Vs	1000	8500	700	1500-2000
Saturated (Peak) Electron Velocity	×10 ⁷ cm/s	1.0 (1.0)	1.3 (2.1)	2.0 (2.0)	1.3 (2.1)
Critical Breakdown Field	MV/cm	0.3	0.4	3.0	3.0
Thermal Conductivity	W/cm K	1.5	0.5	4.5	>1.5
Maximum Operating Temperature	°C	300	300	600	700
RFOM Ratio	NA	1.0	9.6	3.1	24.6
JEM Ratio	NA	1.0	3.5	60	80

According to Table 1-3, the GaN can provide several advantages. They are contributed to the suitability of GaN-based devices for the key applications such as sensing systems, microwave transmitters and power electronics [5]. Power electronics are also the most beneficiaries. High-efficiency power electronics can be produced on AlGaIn/GaN platform mainly because of their favorable device characteristics including high breakdown, low on-resistance, and high switching speed [6] - [9]. Additionally, compare to other wide bandgap competitors, high quality single crystal heterostructures have been successfully obtained and used to fabricate high-performance AlGaIn/GaN devices. The competitive advantages for commercial product due to its excellent properties can be summarized in Table 1-4 [4].

Table 1-2 Competitive advantages for commercial product of GaN devices

Need	Enabling Feature	Performance Advantage
High Power/Unit Width	Wide Bandgap High Field	Compact Ease of Matching
High Voltage	High Breakdown Field	Eliminate/Reduce Step Down
High Linearity	HEMT Topology	Optimum Band Allocation
High Frequency	High Electron Velocity	Bandwidth, μ -wave
High Efficiency	High Operating Voltage	Power Saving, Reduced Cooling
Low Noise	High Gain, High Velocity	High Dynamic Range Receivers
High Temperature Operation	Wide Bandgap	Rugged, Reliable, Reduced Cooling
Technology Leverage	Direct Bandgap, Enable for Lighting	Driving Force for Technology, Low Cost

However, most of these key applications such as high power and high temperature applications are under harsh environment. On-chip monitoring and protection are needed to be taken, especially for high temperature applications, since all semiconductors work well only in a limited temperature range no matter how robust they are. Therefore, temperature sensing technology on GaN-platform is essential for the benefit of on-chip monitoring and protection.

1.2 Introduction of Temperature Sensor

Temperature sensor is a device that can convert temperature changes into readable signals such as voltage. It usually consists of two important elements: the temperature sensor in which some physical change occurs with temperature change, plus some means of converting this physical change into a readable value [10].

Currently, several temperature sensing techniques are in widespread usage. The most common of these are RTDs, thermistors, thermocouples, and sensor ICs. The choice for a specific application depends on the required temperature range, linearity, accuracy, cost, features, ease of designing the necessary support circuitry and some other unique requirements. The common working principle of each technique is that, as temperature varies, certain physical changes occur which can be expressed in some format of electrical characteristic, such as the change of resistance, current, voltage and so on. Subsequently, the most commonly used temperature sensors are mainly introduced in the following.

1.2.1 Resistance Temperature Detector (RTDs)

Resistive sensors use a sensing element whose whole resistance varies with temperature. All integrated resistors have an important thermal sensitivity, but in general they do not show good linearity for highly accurate applications. Although some of these have better linearity such as platinum, they still need correction at extreme temperatures. This usually consists of a coil of platinum wire wound around a bobbin, or a film of platinum deposited on a substrate, characteristic of which is shown in Fig. 1.2.1 [11]. Other drawbacks are that its cost tends to be relatively high, and it is not compatible with some on-chip applications. Also self-heating can occur in some RTDs.

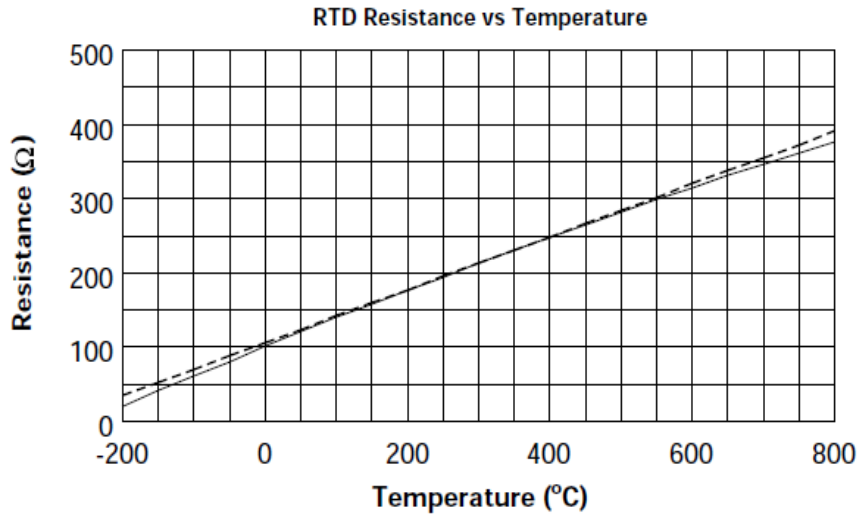
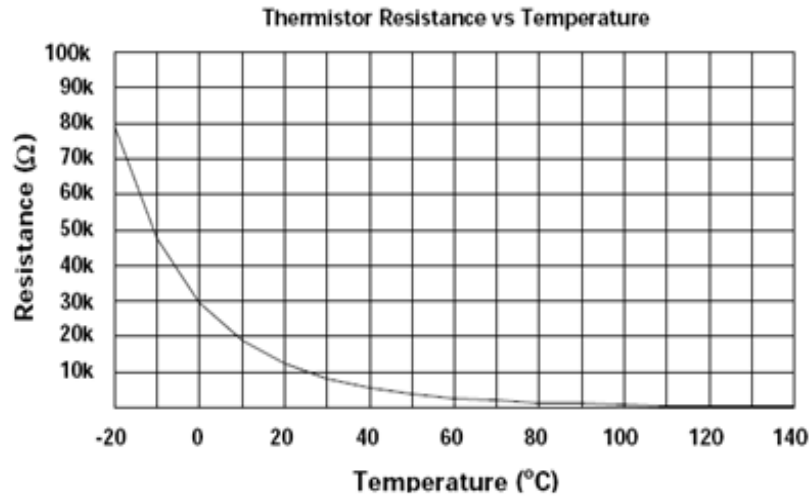


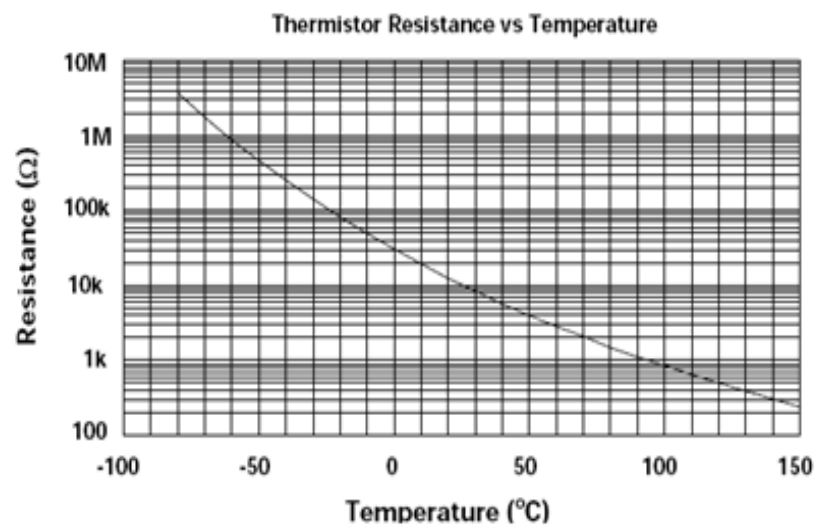
Fig. 1.2.1 Platinum RTD resistance vs. temperature (straight line for reference) [11].

1.2.2 Thermistor

Another type of resistive sensor is thermistor. “Thermistor” is a portmanteau of thermal and resistor. As its name means, it is also a type of resistor with significant temperature dependent resistance. However it differs from RTD. Generally thermistor is made of ceramic or polymer instead of pure metal. Therefore, the cost of thermistor is relatively low. Low-cost thermistors are often used to perform simple measurement or trip-point detection functions in low-cost systems. A thermistor’s resistance-temperature function is very nonlinear (Fig. 1.2.2 [11]), substantial linearization is needed to wide temperature range detection. Additionally, compared to RTD, thermistor only performs within a limited temperature range, typically -90 °C to 140 °C.



(a)



(b)

Fig. 1.2.2 Thermistor resistance vs. temperature:
(a) linear scale. (b) logarithmic scale [11].

1.2.3 Thermocouples

A thermocouple is a junction between two different metals that produces a voltage related to a temperature difference. They can measure a wide range of temperatures. The main limitation is accuracy: system errors of less than one degree Celsius ($^{\circ}\text{C}$) can be difficult to achieve. A principle schematic of thermocouples is shown in Fig. 1.2.3. Any junction of dissimilar metals will produce an electric

potential related to temperature. Thermocouples for practical measurement of temperature are junctions of specific alloys which have a predictable and repeatable relationship between temperature and voltage. Different alloys are used for different temperature ranges. Properties such as resistance to corrosion may also be important when choosing a type of thermocouple.

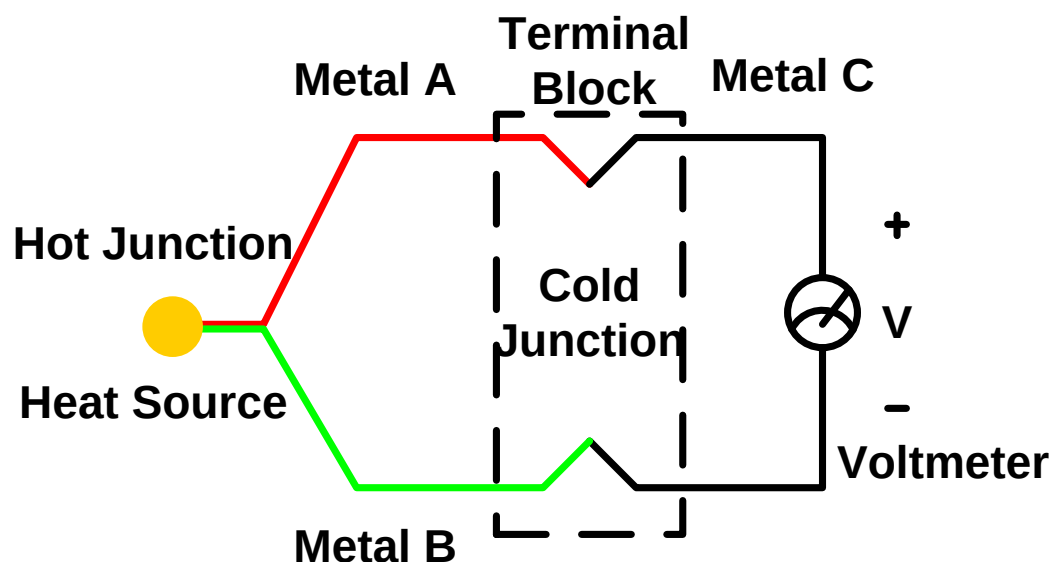


Fig. 1.2.3 Schematic of thermocouple.

Thermocouple is widely used partly because of its low thermal mass and very wide temperature range which can be up to 1700 °C. Table 1-2 shows Seebeck coefficients and temperature ranges for several types of thermocouples. They are usually used in temperature measurement for kilns, gas turbine exhaust, diesel-engines, and other industrial processes.

Table 1-3 Seebeck coefficient and temperature range for several thermocouple types

Type	Junction Material	Seebeck Coefficient ($\mu\text{V}/^\circ\text{C}$)	Temperature Range ($^\circ\text{C}$)
E	Chromel-Constantan	58.5@ 0 $^\circ\text{C}$	0 to 1700
J	Iron-Constantan	50.2@ 0 $^\circ\text{C}$	0 to 750

K	Chromel-Alumel	39.4@ 0°C	-200 to 1250
R	Platinum-Rhodium	11.5@ 0°C	0 to 1450

However, thermocouple has several disadvantages. The main drawback is the small sensitivity – in the order of tens of microvolt per degree Celsius. On this level of sensitivity, it calls for a low-offset amplifier to produce a usable output voltage. Nonlinearity of temperature-to-voltage transfer function is another drawback. It amounts to many degrees of variation within the whole operation range. To make up this disadvantage, a compensation circuit or look-up table is always necessary. There are still some technical problems when applying thermal couplers. When the measurement point is far from the measuring instrument, the intermediate connection needs to be made by extension wires which are less costly than the materials used to make the sensor. Since thermocouples usually need to be standardized against a reference temperature of zero degrees Celsius, complex support methods are required to assist it; practical instruments use electronic methods of cold-junction compensation to adjust the varying temperature at the instrument terminals.

1.2.4 Sensor ICs on Silicon-based Technology

So far, the most commonly used temperature sensor ICs are those on silicon platform. It has already been a well developed technology. The nominal operating range of silicon-based temperature sensors is -55 °C to +150 °C. This kind of sensor can integrate several functional blocks into one package which, to a great extent, simplifies the broadest ranges of temperature sensing challenges. Extra compensation, linearization, and comparators can be eliminated unless special applications demanded.

A diagram of the integrated temperature sensor is shown in Fig. 1.1.2 [12]. This sensor integrated digital temperature sensor and accurate A/D convertor.

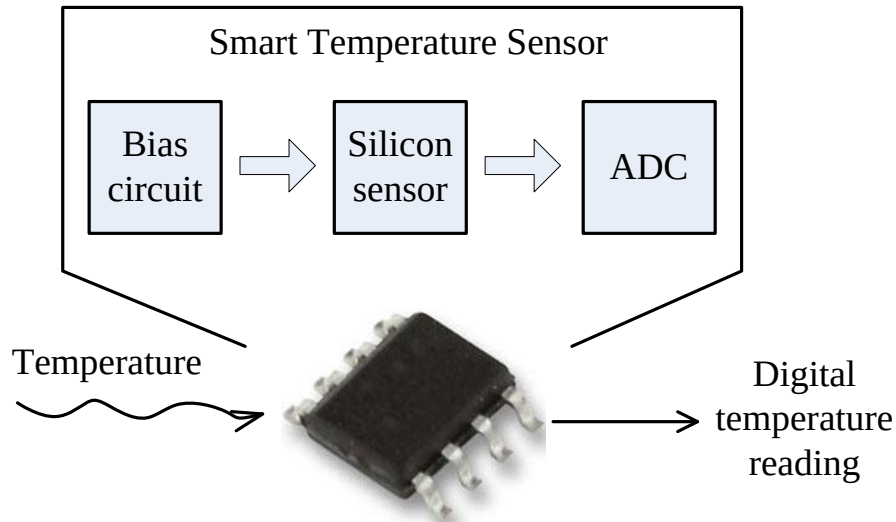


Fig. 1.2.4 Diagram of a silicon-based integrated smart temperature sensor [12].

A wide variety of silicon-based sensors are proposed. Some of these are analog circuits, with either voltage or current output. Others combine analog sensing circuitry with digital I/O and control registers, making them an ideal solution for microprocessor-based applications.

The main sensing components on the silicon-based technology are PN junction of the bipolar transistors which also take the fabrication-compatible into consideration. The bipolar transistor base-emitter junction is a PN junction whose junction potential (V_{be}) have a property that can be exploited to produce a voltage that is changed to temperature. The combination of two such voltages yields a voltage that is "proportional to the absolute temperature" (PTAT). If the transistor and the sensing interface circuits are optimized to minimize other effects which depend on temperature, the voltage will be a highly accurate representation of the temperature over a wide temperature range. As fabrication compatibility challenges are taken into

consideration, bipolar devices are then preferred as thermally sensitive elements, therefore most of the temperature sensor ICs have been traditionally designed using bipolar technologies. Subsequently, because of the additional advantages of the CMOS technology, many of those designs have migrated from bipolar to CMOS technologies, which also use the parasitic PNP bipolar transistor in CMOS technology.

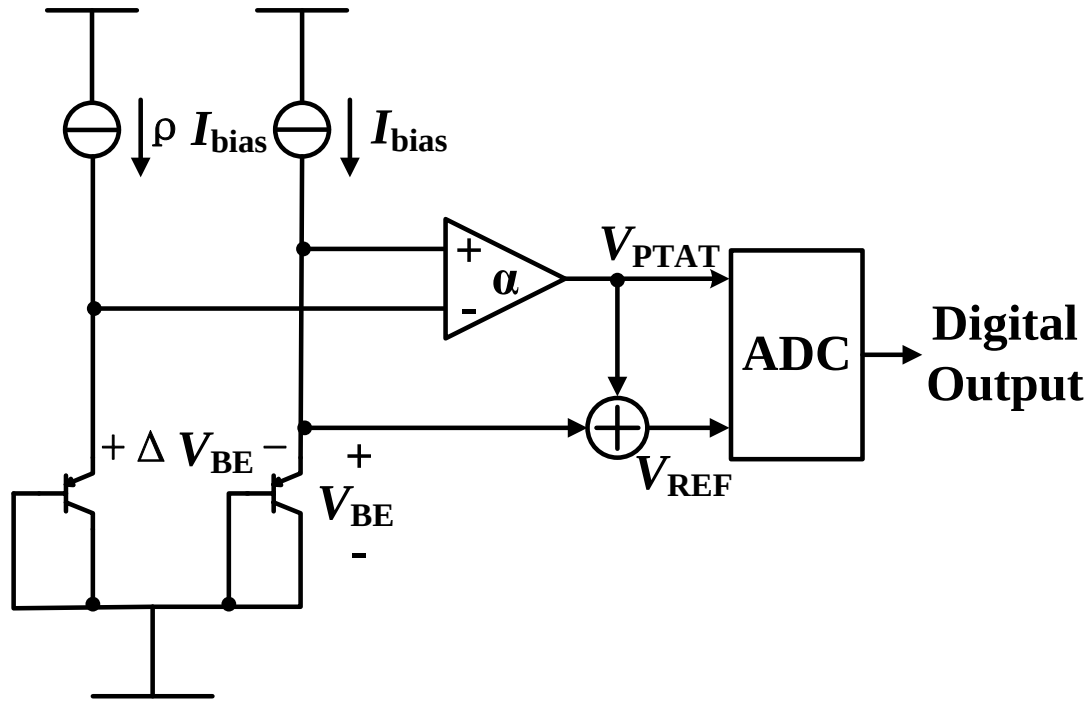


Fig. 1.2.5 Schematic of temperature sensor IC on silicon-based technology.

Fig. 1.2.4 shows the schematic of temperature sensor ICs on silicon-based technology [13][14]. The heart of the sensor consists of two diode-connected PNP bipolar transistors which are biased at different current densities. They are used to produce two voltages V_{BE} and ΔV_{BE} . The produced ΔV_{BE} is proportional to absolute temperature, which is known as PTAT, and the combination of the two voltages mentioned above is used to produce a reference voltage. Then, through an analog to digital converter (ADC), the PTAT voltage can be converted to a digital temperature reading output.

Although silicon-based temperature sensing technology has already been well developed, on GaN platform, we cannot just copy the silicon-based design for GaN-based on-chip temperature sensing, since we lack of high quality P-type material due to the low active efficiency of holes in AlGaIn/GaN material. On GaN-based platform, high electron mobility transistors (HEMT) are used instead of MOSFET, and Schottky barrier diode (SBD) in addition to lateral field-effect rectifier are applied instead of PN junction. The following chapter will introduce them respectively in details.

1.3 GaN-based Platform

On GaN-based platform, there are HEMTs, SBD and L-FER that can be used to temperature sensor design.

1.3.1 AlGaIn/GaN HEMT

AlGaIn/GaN HEMTs can be divided into two types, one is depletion-mode (D-mode) HEMT, which is normally-on, the other is Enhancement-mode (E-mode) HEMT which is normally-off. The cross-section of these two types of HEMTs is shown in Fig. 1.3.1.

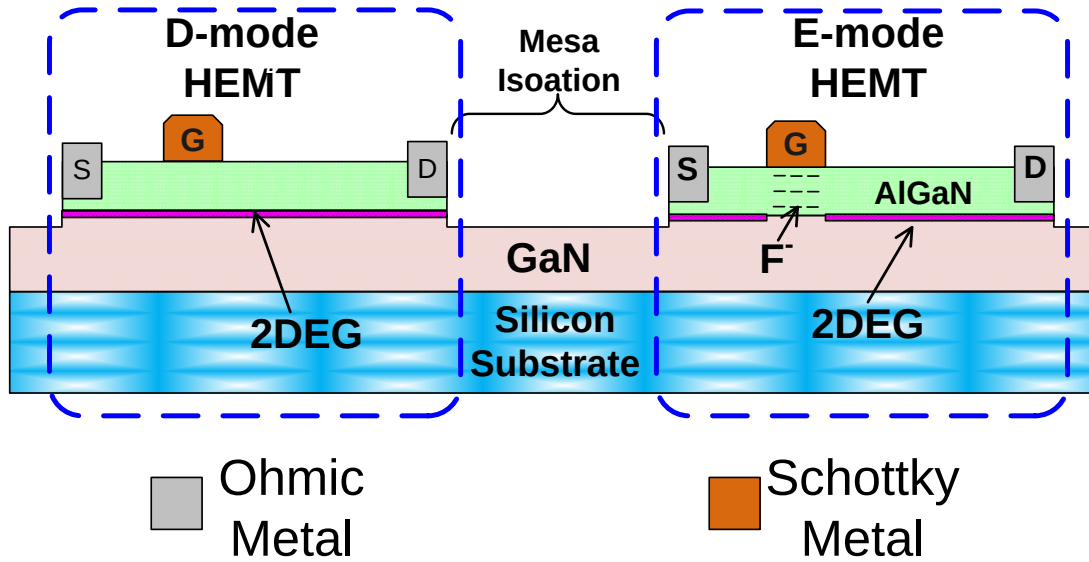


Fig. 1.3.1 Cross-section of D-mode HEMT and E-mode HEMT

From the cross-section of these two HEMTs, we can see the difference between them is the electron depletion in the two-dimensional electron gas (2DEG) channel under the gate region of E-mode HEMT. This is due to the fluorine ion plasma implantation which depletes the electron in 2DEG channel to shift the threshold voltage from a negative value to positive value. In this way, a normally-off device is formed.

1.3.2 AlGaN/GaN SBD

Due to the shortage of high-quality P-type material, a Ni/Au Schottky contacts were deposited on the top of AlGaN layer to form a vertical SBD. The cross-section and energy-band diagram of SBD is shown in Fig. 1.3.2. There are lots of literatures researched on SBD and its application, its long-term stability is still an issue to be concerned for large current applications. Additionally, from the energy-band diagram of SBD, we can see a heterojunction in series with Schottky junction. This makes SBD high turn-on voltage and large on-resistance, which are not the expected characteristics of an ideal diode. However, all these issues would not be a big concern

in the sensor design, since in the sensor proposed in this work apply sub-threshold region of SBD.

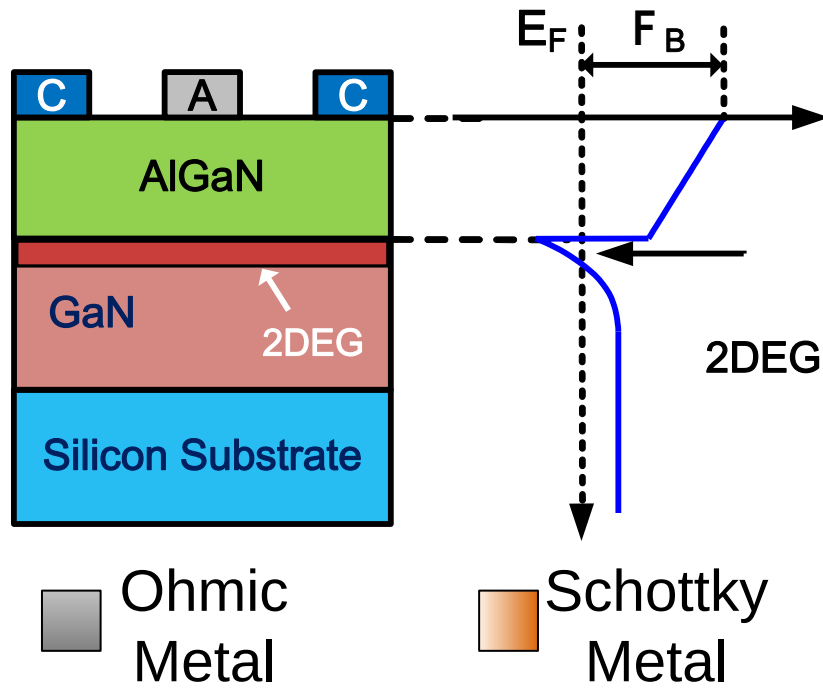


Fig. 1.3.2 Cross-section of SBD

1.3.3 AlGaN/GaN L-FER

To avoid the issues of the SBD, another kind of rectifier is proposed by our group recently, namely later field-effect rectifier (L-FER). L-FER is also a normally-off device. It features a schottky-gate-controlled 2DEG channel which lies between the anode and cathode. By tying up the Schottky gate and anode together, the forward turn-on voltage is determined by the threshold voltage of channel instead of schottky barrier. Therefore, the long-term stability is effectively improved and the forward turn-on voltage as well as on-resistance is reduced compared to SBD [25].

The cross-section of L-FER is shown in Fig. 1.3.3. When it is forward-biased, the electrons in 2DEG channel under the gate region are induced, and the L-FER is on. When it is reverse-biased, the electrons in 2DEG channel under the gate region are depleted, and the L-FER is off.

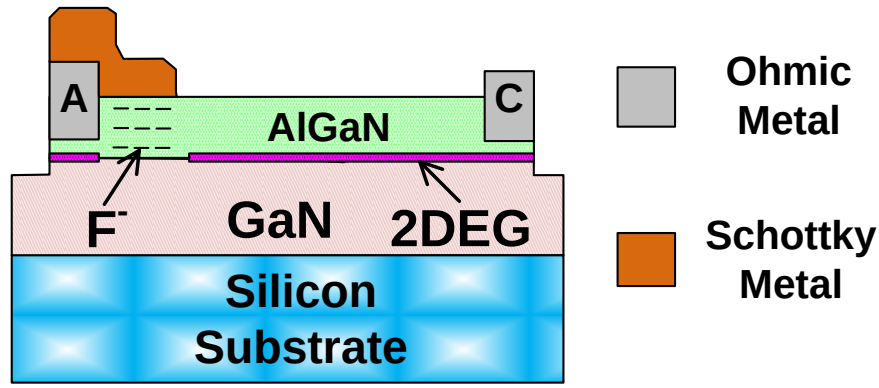


Fig. 1.3.3 Cross-section of L-FER

1.4 Motivation of This Work

To optimize the performance of integrated circuits on GaN-based platform, it is essential to build a temperature sensing circuit which can be compatible to GaN-based device technology since a monolithically integrated solution is preferred to achieve low cost, high reliability and high integration. Although integrated temperature sensing ICs on silicon have been successfully developed [15][16], the maximum operation temperature of the conventional silicon is 200 °C. However, the sensing elements' should exhibit more robust temperature tolerance than target operational blocks to satisfy an effective temperature detection and protection. Additionally, CMOS technology is not compatible with GaN-based HEMT technology, thus, silicon-based temperature sensing IC is not an excellent candidate on GaN platform. It is indispensable to propose a temperature sensing circuit as well as over-temperature protection using HEMT, SBD or LFER which is HEMT compatible on GaN-based technology.

The GaN-based platform is illustrated in Fig. 1.4.1, which integrated the power devices and peripheral devices on a single GaN platform. AlGaN/GaN devices have been developed for over 20 years and they have already matured for the design of micro-wave power amplifiers and high-voltage power switches in military

applications. Fig. 1.4.2 shows the essential smart power functional blocks, and most of these have been demonstrated on GaN platform. However, there is a big gap in the context of temperature sensing and control circuits on this platform. Therefore, the exploration to design temperature sensing and over-temperature protection ICs are imperative. This thesis reports three main works that are 1) a GaN-based temperature sensor, 2) a PTAT sensing IC, and 3) an over-temperature protection for smart power technology.

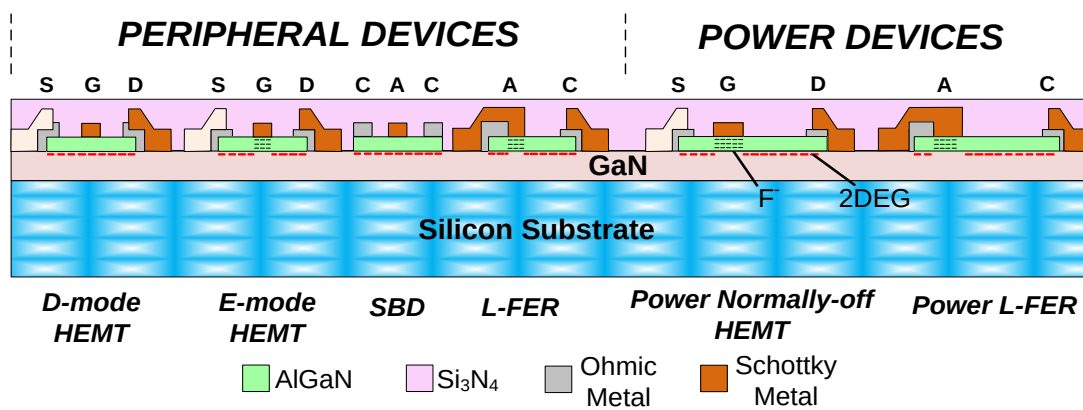


Fig 1.4.1. GaN-based power electronic platform.

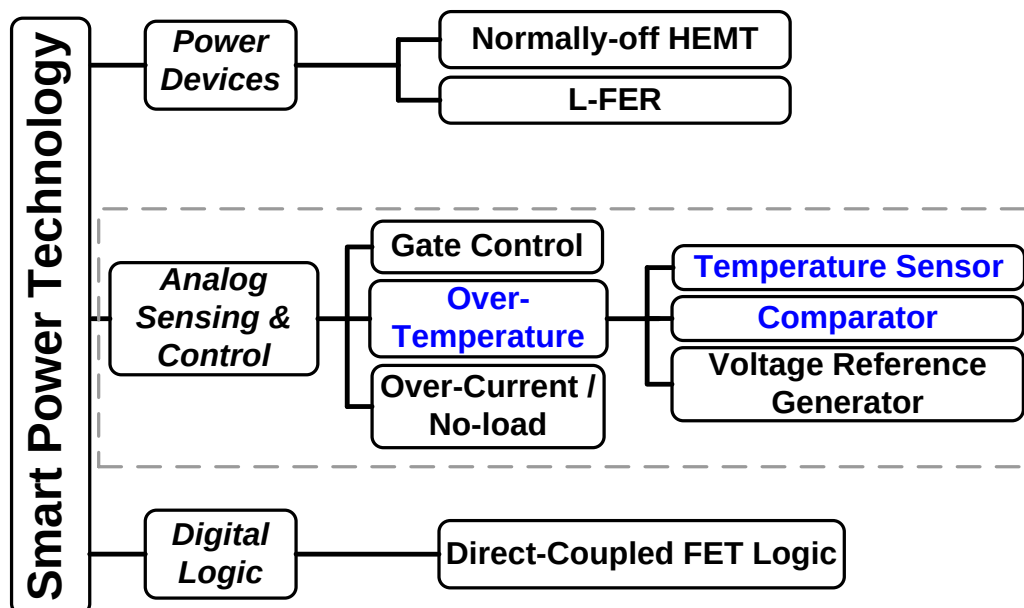


Fig 1.4.2. A diagram of smart power functional blocks.

1.5 Objective of This Work

This work mainly focuses on the GaN-based temperature sensing and over-temperature protection circuit. A GaN-based temperature sensor is demonstrated, whose simple structure facilitates the monolithic integration with other functional blocks. Then a PTAT temperature sensor is developed, which shows better linearity and accuracy. Finally, based on the design of GaN-based temperature sensor, a previous designed single polarity power supply bootstrapped comparator is coordinated to implement an over-temperature protection for smart power IC on GaN platform.

Chapter 2 investigates the temperature dependence of AlGaIn/GaN HEMTs, AlGaIn/GaN SBDs, and HEMT-compatible L-FERs on AlGaIn/GaN heterostructure wafer on silicon substrate. The temperature-dependent current-voltage characteristics are summarized for these devices to lay a foundation for temperature sensing circuit design.

Chapter 3 reports a novel design of the temperature sensing technology on the GaN-based platform. The GaN-based temperature sensor is designed with AlGaIn/GaN HEMT and LFERs, which avoid complicated structures. It converts temperature changes into output voltage which can be easily applied by electronic blocks such as ADC or comparator in an over-temperature protection. A stable negative temperature coefficient is achieved. In addition, power supply rejection ratio (PSRR) with varied frequency at room temperature and temperature-dependent PSRR at single temperature are also reported.

Chapter 4 introduces a proposed GaN-based PTAT temperature sensor, which is derived from the bandgap working principle on silicon-based temperature sensing technology. In order to achieve GaN-based devices compatibility, the AlGaIn/GaN

heterostructure Schottky barrier diode (SBD) is applied instead of PN junction. A current mirror is also demonstrated using all-D-HEMT in place of P-type devices to ensure the same current flowing into two sensing SBDs. A stable positive temperature coefficient is presented with better linearity and accuracy. Subsequently, PSRRs and temperature-dependent PSRR are also discussed which is obviously better than the previous one.

Chapter 5 presents a design of over-temperature protection for smart power technology on GaN platform. It is designed to provide a protection for the power device in the smart power management system. The over-temperature protection consists of the GaN-based temperature sensor and bootstrapped comparator. When temperature rises above the specific point, a sharp shut down voltage will be provided by over-temperature protection. However when temperature is below the specific value, the over-temperature protection will have no influence on the target power device. A changeable pre-determined specific temperature point can be achieved through changing the reference voltage of comparator, the result of which is also presented subsequently.

Finally, the work is summarized in Chapter 6, and future plan for the thesis work is also suggested.

Chapter 2 Temperature-dependent Characteristic of GaN-based Devices

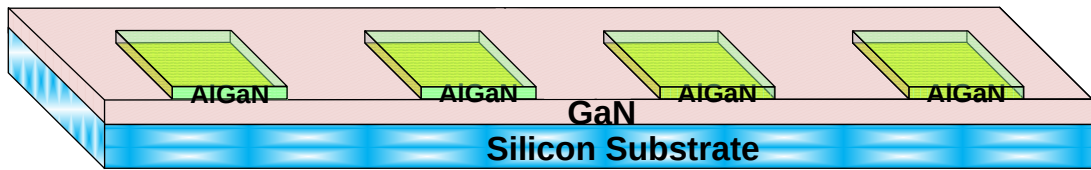
2.1 Motivation

In this chapter, temperature dependent characteristics of GaN-based devices are investigated in order to facilitate temperature sensor designs. The sensors are designed by the right of the unique temperature characteristics of these wide bandgap devices including D-mode HEMT, E-mode HEMT, L-REF, and SBD. As a result, AlGaIn/GaN heterostructure devices not only can deliver a robust performance under extreme high temperature, but also own strong temperature dependent current density after the device is turned on [19] - [22]. The strong temperature dependent current density of GaN-based devices is attributed to the reduction of the 2DEG mobility which is caused by stronger phonon scattering at higher temperature. However the threshold voltage of 2DEG channel shows a stable thermal characteristic. Thus the temperature sensor designs benefit from strong temperature dependent current density as well as stable operation under high temperature condition greatly [23].

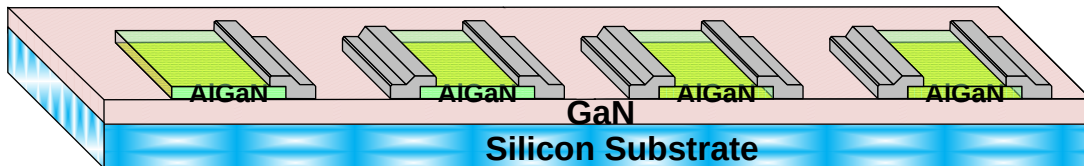
2.2 Fabrication

All monitor devices are fabricated on commercially available $\text{Al}_{0.26}\text{Ga}_{0.74}\text{N}/\text{GaN}$ HEMT wafer grown by metal-organic chemical vapor deposition (MOCVD) on a 4-inch (111) silicon substrate. The epitaxial structure includes a 1.8 μm -thick GaN buffer, a thin AlN interface enhancement layer, 18 nm undoped AlGaIn barrier and a 2 nm GaN cap layer. The starting wafer featured a sheet resistance of 330 ohm/square, 2DEG mobility of $1800 \text{ cm}^2/\text{V}\cdot\text{s}$ and a pinch off voltage of -2.1 V.

The fabrication process is shown in Fig. 2.2.1. It starts with the mesa etching isolation. It used inductively coupled plasma reactive ion (Cl_2/He) dry etching (ICP-RIE). Then, e-beam sequence of Ti/Al/Ni/Au (20 nm/150 nm/50 nm/80 nm) was used to form ohmic contacts and subsequent rapid thermal annealing (RTA) at 850 °C for 30 s. After annealing recipe, the ohmic contact resistance was measured as 0.85 $\Omega\text{-mm}$ by a standard transfer length method (TLM) procedure. Afterwards, the Schottky contacts of E-mode devices and L-FERs were defined by photolithography and the subsequent CF_4 plasma ion implantation at 110 W for 170 s. To repair the plasma-induced damage in the AlGaN barrier the sample was annealed at 400 °C for 10 minutes. The negatively charged fluorine ions effectively depleted the 2DEG channel and shifted the threshold voltage from the original -2.1 V to +0.2 V. Subsequently, the self-aligned Schottky contacts were deposited by e-beam evaporation of Ni/Au and liftoff to form E-mode devices and L-FERs. Finally, the second Schottky contacts were defined and deposited to realize D-mode HEMTs and SBDs.



(a) Mesa isolation



(b) Ohmic contacts

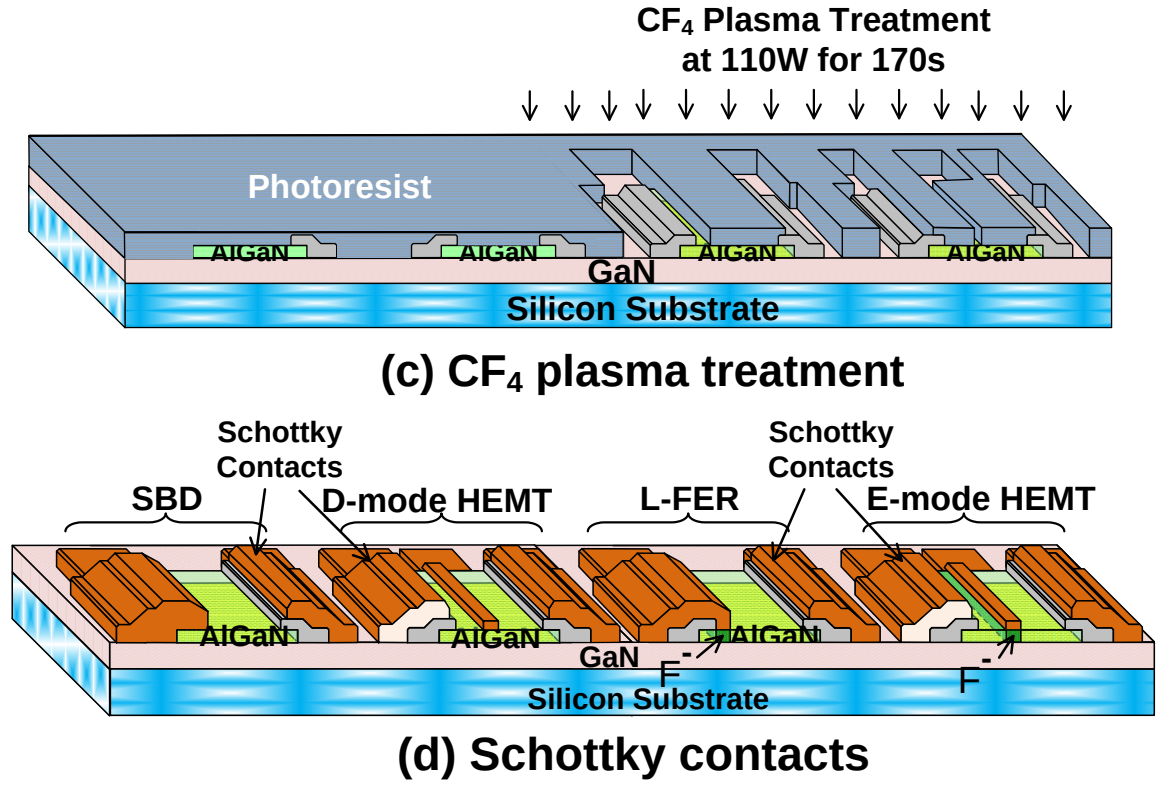


Fig. 2.2.1 Process flow of GaN-based devices fabrication

2.3 Temperature-dependent Characterization

All D-mode HEMT, E-mode HEMT, L-FER and SBD are characterized by Agilent 4142B modular DC source/monitor and SIGNATONE S-1060 probe station equipped with a temperature hot chuck system. The measurement results are discussed as follows respectively.

2.3.1 D-mode HEMT

The temperature dependent I-V characteristics for a D-mode HEMT with 2 μm gate length, 1.5 μm for both gate-to-source and gate-to-drain distance, are shown in Fig. 2.3.1. At room temperature (RT), D-mode HEMT exhibit a maximum drain current density of 325mA/mm at $V_{GS} = 0$ V with a threshold voltage of -2.1 V. As V_{DS} sweeps up, the drain current decreases slightly mainly because of the device's self-heating. At 250 $^{\circ}\text{C}$, due to severer phonon scattering induced electron mobility

reduction in 2DEG channel, the saturation drain current decreases by more than half of the original current. Since the current decreases as temperature rises, the transconductance also decreases which is measured with $V_{DS} = 10\text{V}$. However, the threshold voltage exhibits thermal stability.

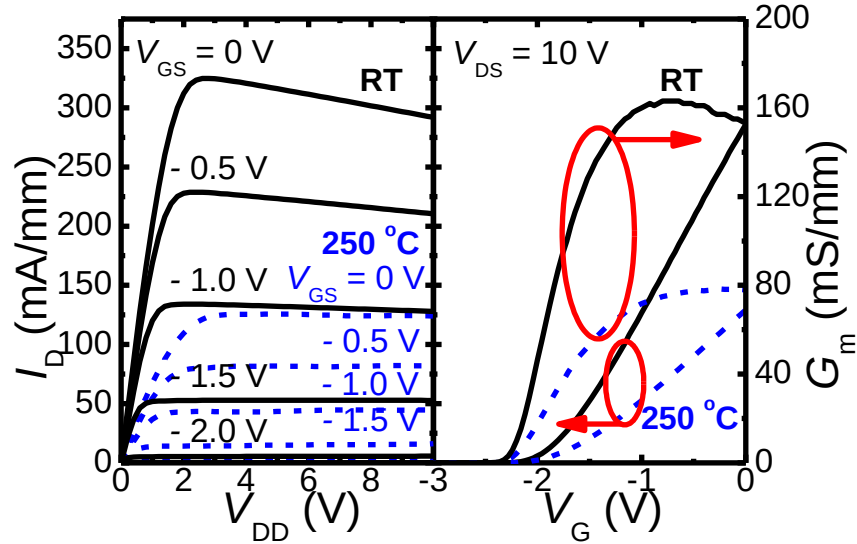


Fig. 2.3.1 Temperature dependent I - V characteristic of D-mode HEMT.

2.3.2 E-mode HEMT

The temperature dependent I - V characteristics of E-mode HEMT are plotted in Fig. 2.3.2. The E-mode HEMT is fabricated with fluorine treatment to shift linear threshold voltage from original -2.1 V to 1.05 V . The size of the measured E-mode HEMT is $2\text{ }\mu\text{m}$ gate-length, $1.5\text{ }\mu\text{m}$ for both gate-to-source and gate-to-drain distance. At room temperature, the maximum drain current density is 280 mA/mm at $V_{GS} = 3\text{ V}$. Again, the decreased saturation drain current occurs when V_{DS} increases, which is due to device's self-heating. As temperature rises up to $250\text{ }^{\circ}\text{C}$, due to electron mobility which is also caused by stronger phonon scattering, the drain current density decreases to 167 mA/mm . However, compared to D-mode HEMT, it shows less temperature dependence, it is because the implanted fluorine plasma caused impurity

scattering also accounts for the scattering mechanism of the E-mode HEMT, which depends little on temperature.

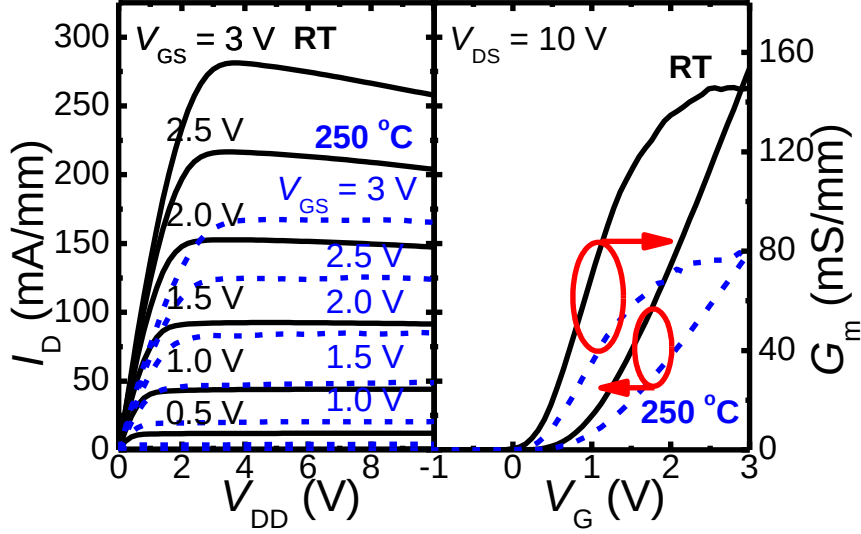


Fig. 2.3.2 Temperature dependent I - V characteristic of E-mode HEMT.

2.3.4 Schottky Barrier Diode

Because of lack of high-quality p-type material, Ni/Au Schottky contacts are deposited on the top of AlGaN/GaN to form a vertical Schottky rectifier. However due to the unipolar nature of the GaN-based SBD, it is endowed with faster switching speed and less reverse recovery current compared to the p-n junction diode [24]. The temperature dependent I - V characteristic is presented in Fig. 2.3.3.

It can be seen that SBD has a strong temperature dependent I - V characteristic. However, since the dominating mechanism of operation is different, different temperature coefficient occurs at a critical voltage point of 1.5 V. When the forward bias voltage is less than 1.5 V, SBD mechanism dominates the operation, and the I - V relationship shows a positive temperature coefficient. When the forward bias voltage is larger than 1.5 V, the parasitic resistor dominates the operation, and the SBD shows a negative temperature coefficient.

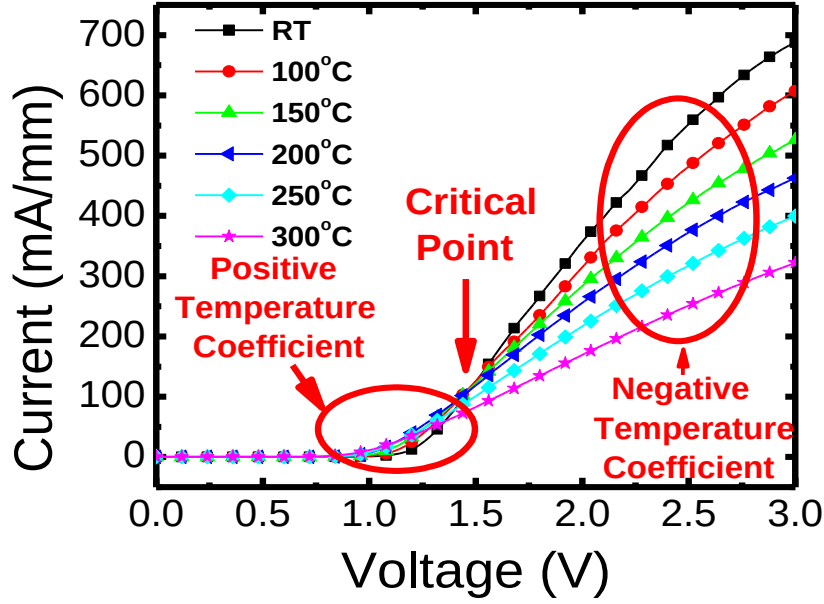


Fig. 2.3.3 Temperature dependent I - V characteristic of SBD.

2.3.4 Lateral-field Effect Rectifier

The monitored LFERs of this work are fabricated with the same process with E-mode HEMT, therefore a fluorine treatment is applied to L-FER to form a thermal stable positive turn-on voltage. Both forward and reverse I - V characteristics are illustrated in Fig. 2.3.5. The turn-on voltage is measured as 0.3 V. The forward biased current of L-FER shows a negative temperature dependent, since it decreases as a function of rising temperature, but it is not as strong as D-HEMT, when the voltage across L-FER is less than 3V. Therefore, when L-FER is connected to a zero biased D-HEMT, different temperature coefficients appear, which can lead to a temperature dependent voltage across L-FER. The size of measured L-FER is with 2 μm Schottky gate length and 5 μm drift length.

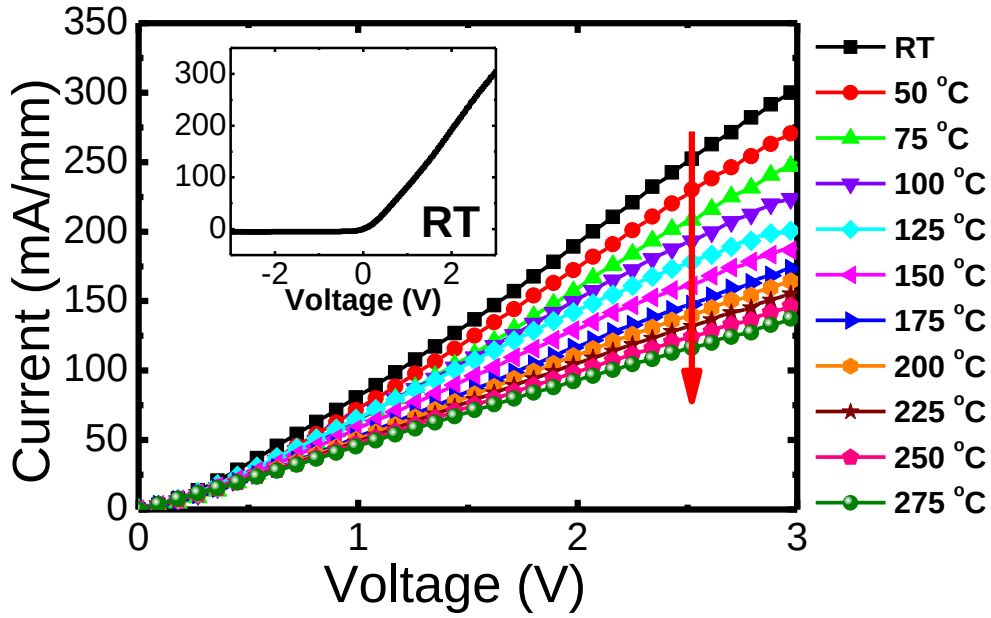


Fig 2.3.5 Temperature dependent I - V characteristic of L-FER.

2.4 Summary

The temperature dependent I - V characteristics of GaN-based devices including D-mode HEMT, E-mode HEMT, SBD and LFER are discussed in this chapter. All of these devices strongly depend on temperature to different extent. Due to the unique properties of GaN material and AlGaIn/GaN heterostructure devices, they have been demonstrated to deliver an outstanding performance under high-temperature condition with strong stability and reliability. The results lay a solid foundation to facilitate the design of temperature sensor and over-temperature protection on GaN platform.

Chapter 3 GaN-based temperature sensor

3.1 Motivation

The wide-bandgap GaN semiconductors, with wide operating temperature ranges, are capable of delivering super performance under some extreme harsh conditions [25]-[27]. High-quality HEMTs and L-FERs have been obtained and used to fabricate high-performance ICs [28]-[31]. Temperature sensing for the on-chip detection purpose is imperative for performance optimization of GaN-based integrated circuits. Since intrinsic carrier mobility is influenced by phonon scattering, which is temperature-sensitive, these devices exhibit strong temperature dependences in different degree. Thus, GaN-based temperature sensor using active D-mode HEMT and L-FERs on AlGaN/GaN heterostructures not only can conduct a wide monitor range with high stability, but also potentially lead to integrated over-temperature protection solutions for on-chip thermal detectors and temperature controllers.

3.2 Design of GaN-based temperature sensor

The GaN-based temperature sensor is implemented with a gate-source connected D-HEMT (normally-on) which acts current source and two L-FERs (lateral field-effect rectifiers). The design of the proposed sensor is shown in Fig 3.2.1. The dimensions of M1 are chosen to be 2 μm long and 5 μm wide to enable smaller process variation. The two L-FERs gates dimension were identically set as 1.5 μm long and 40 μm wide, for smaller on resistance and proper sensing output voltage for the next stage application.

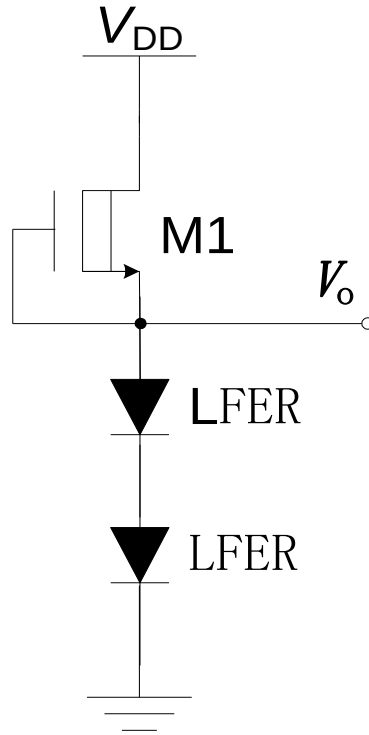


Fig. 3.2.1 Design of GaN-based temperature sensor.

The sensing principle of this design is based on the different temperature-dependent characteristics of D-mode HEMT and LFERs, which is shown in Fig. 3.2.2. As temperature rises to 275 °C, the current flowing through M1 decrease more than half of its original value, however, the current of LFERs is not supposed to decrease so largely, it means the current of LFER is smaller than it should be at 275 °C. Therefore the significant decrease in current of LFER forces the voltage across LFER decreasing, which leads a negative temperature coefficient at output voltage. The two series-connected L-FERs structure is to enlarge the output voltage compared to only one L-FER, however the minimum supply voltage also increase, and the minimum power consumption needs to be sacrificed.

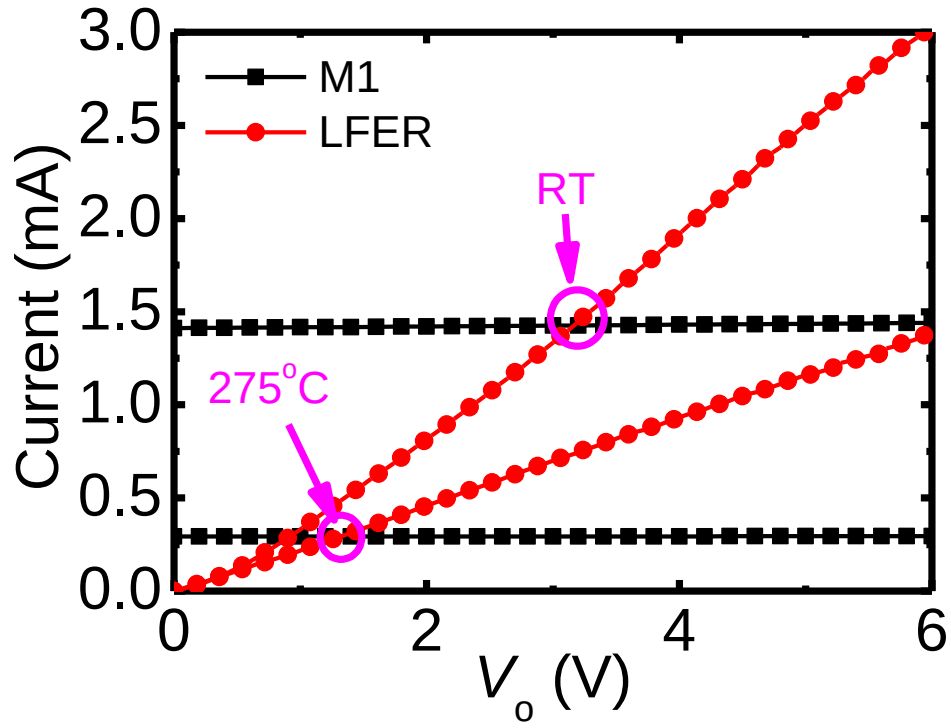
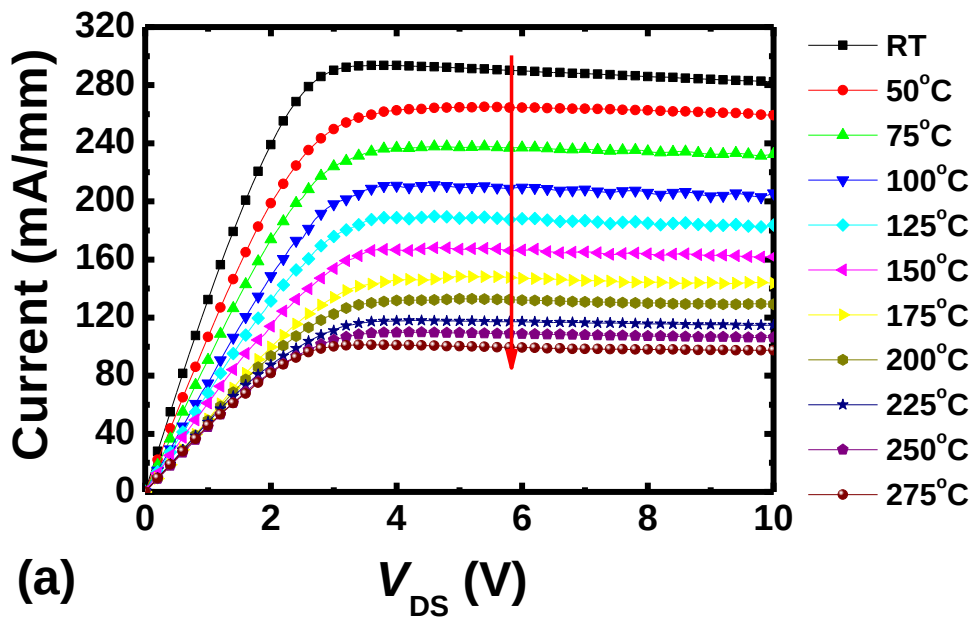


Fig. 3.2.2 Operating principle of GaN-based temperature sensor

Fig. 3.2.3 characterizes the I - V properties of the current source (a) and L-FER (b) in detail up to 275 °C respectively.



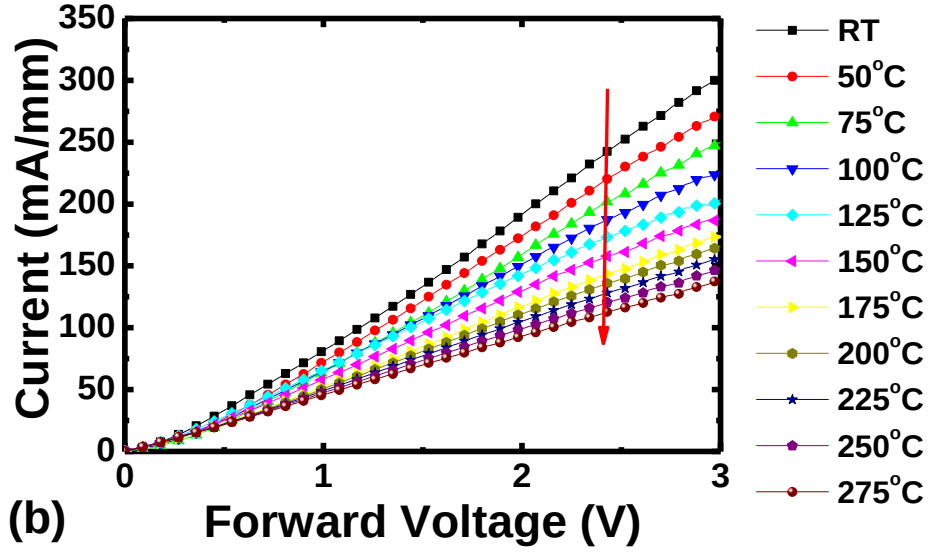


Fig. 3.2.3 Temperature-dependent characteristics of (a) M1 (b) LFERs.

3.3 Fabrication

The sample used to fabricate the designed sensor was a commercially available $\text{Al}_{0.26}\text{Ga}_{0.74}\text{N}/\text{GaN}$ HEMT wafer grown by metal-organic chemical vapor deposition (MOCVD) on a 4-inch (111) silicon substrate. The epitaxial structure includes a 1.8 μm -thick GaN buffer, a thin AlN interface enhancement layer, 18 nm undoped AlGaIn barrier and a 2 nm GaN cap layer. The starting wafer featured a sheet resistance of 330 Ω/square , 2DEG mobility of $1800 \text{ cm}^2/\text{V}\cdot\text{s}$ and a pinch off voltage of -2.1 V.

The micrograph taken by SEM is shown in Fig. 3.3.1. The fabrication process can be illustrated as the following. Firstly, mesa isolation was carried out by inductively coupled plasma reactive ion (Cl_2/He) dry etching (ICP-RIE). Second, the ohmic contacts were deposited by e-beam sequence of Ti/Al/Ni/Au (20 nm/150 nm/50 nm/80 nm) and subsequent rapid thermal annealing (RTA) at 850°C for 30 s. The formed ohmic contact resistance was $0.85 \Omega\cdot\text{mm}$ which is obtained by measuring a standard transfer length method (TLM) procedure. Third, the Schottky contacts of E-mode devices and L-FERs were defined by photolithography and the subsequent CF_4 plasma ion implantation at 110 W for 170 s. Then, the sample was annealed at

400 °C for 10 min to repair the plasma-induced damage in the AlGaN barrier. The negatively charged fluorine ions effectively depleted the 2DEG channel and shifted the threshold voltage from the original -2.1 V to +1.05 V. Then, the self-aligned Schottky contacts were deposited by e-beam evaporation of Ni/Au and liftoff to form E-mode devices and L-FERs. Finally, the second Schottky contacts were defined and deposited to form D-mode devices.

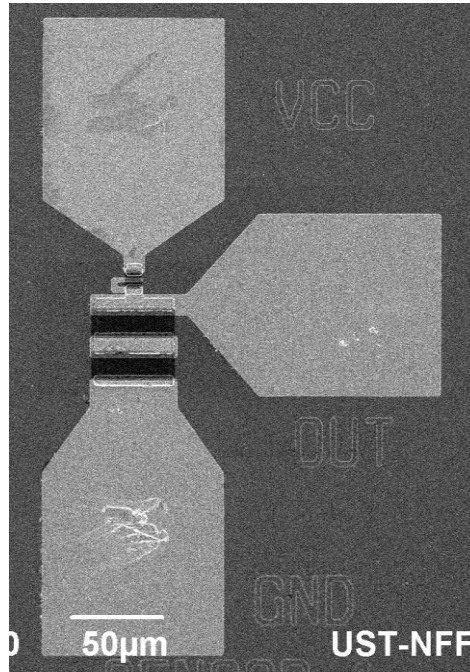


Fig. 3.3.1 SEM image of GaN-based temperature sensor.

3.4 Measurement

3.4.1 Temperature Dependent DC Measurement

The output voltage of the proposed temperature sensor is presented in Fig. 3.4.1. The testing conditions for the sensor are when the supply voltage (V_{DD}) sweeps from 0 V to 15 V and the temperature ranges from RT to 275 °C with a 25 °C increasing per step. Therefore the supply voltage should be no less than 8 V to work properly.

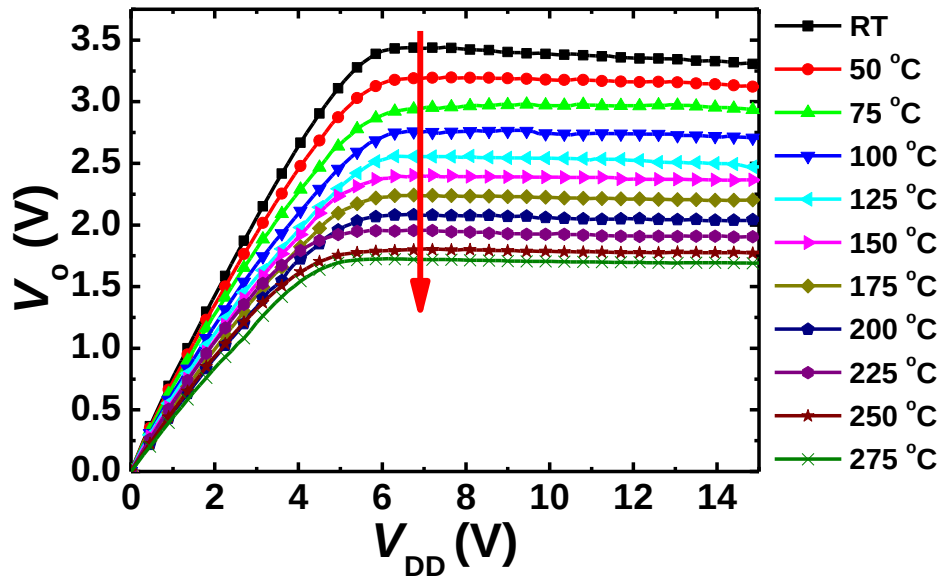
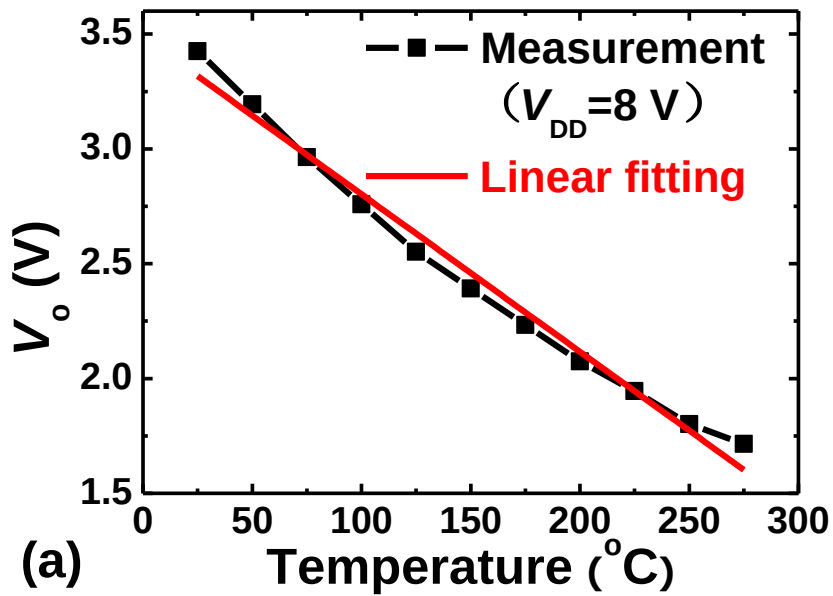


Fig. 3.4.1 Temperature dependent V_o - V_{DD} relationship.

The V_o -T characteristics with different power supply voltages can be extracted. Here we analyze the characteristics of V_o -T as $V_{DD}=8$ V and $V_{DD}=12$ V respectively (Fig. 3.4.2).



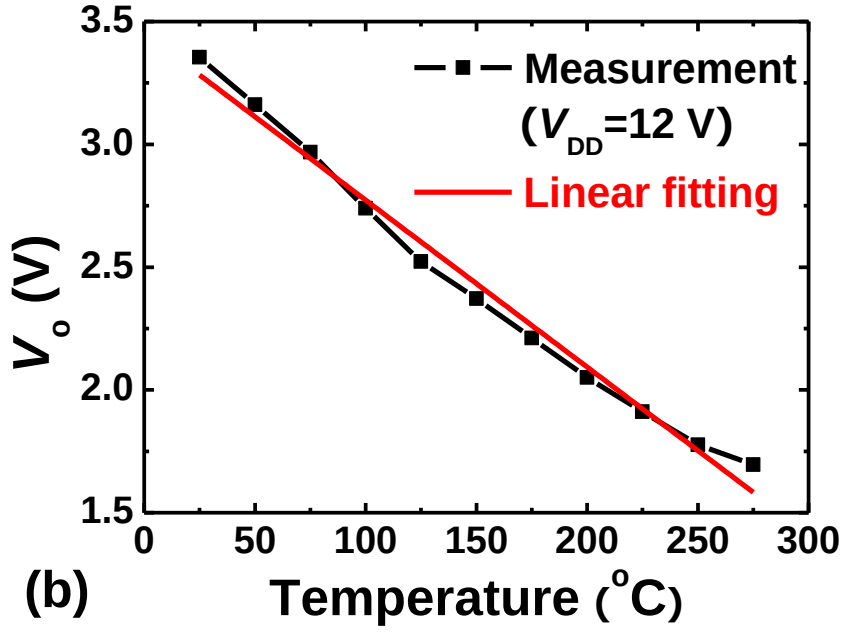


Fig. 3.4.2 Extracted V_o -T characteristics of temperature sensor (a) $V_{DD} = 8$ V; (b)

$$V_{DD} = 12 \text{ V}$$

As shown in Fig. 3.4.2 (a), when $V_{DD} = 8$ V, the output voltage shifts from 3.425 V to 1.715 V as temperature rises from RT to 275 °C. The linear fitting is also presented as the red line. The total voltage drift is 1.71 V from when temperature rises from RT to 275 °C, the temperature coefficient is -6.8 mV/°C, and the output deviation is 0.27 mV/°C, which is 3.96% compare to the output voltage change per °C. An average error is 0.04 °C/mV.

As shown in Fig. 3.4.2 (b), when $V_{DD} = 12$ V, the output voltage shifts from 3.354 V to 1.695 V as temperature rises from RT to 275 °C. The linear fitting analysis result is also shown as the red line. It can be seen that the total voltage drift is 1.66 V when temperature rises from RT to 275 °C, the temperature coefficient is -6.8 mV/°C, and the output deviation is 0.245 mV/°C, which is 3.60% compare to the output voltage change per °C. An average error is 0.04°C/mV.

Table 3-1 The characteristics of V_o - T with different V_{DD}

V_{DD}	Voltage Drift (V)	Temperature Coefficient (mV/°C)	Average Error (°C/mV)
8 V	1.71	-6.8	0.04
12 V	1.66	-6.8	0.04

The characteristics of V_o -T with different V_{DD} can be summarized in Table 3-1. The proposed sensor can deliver a stable negative temperature coefficient up to 275 °C with an average error of 0.04 °C/mV.

3.4.2 Power Supply Rejection Ratio (PSRR)

In monolithic applications such as thermal detecting and over-temperature protection for power electronics, there are several kinds of noises which can have a bad effect on the performance of the temperature sensor, which may come from the digital circuits and switch-mode power converters. To generate a stable temperature dependent output voltage, the temperature sensor should exhibit a reliable performance in a harsh supply noise environment. The figure of merit, PSRR, can express the ability to reject noise from power supply. The PSRR is measured by applying an ac source to power supply and monitoring the output voltage, which is shown in Fig. 3.4.3. The PSRR can be achieved by comparing the voltage change of the output with the voltage change in supply voltage.

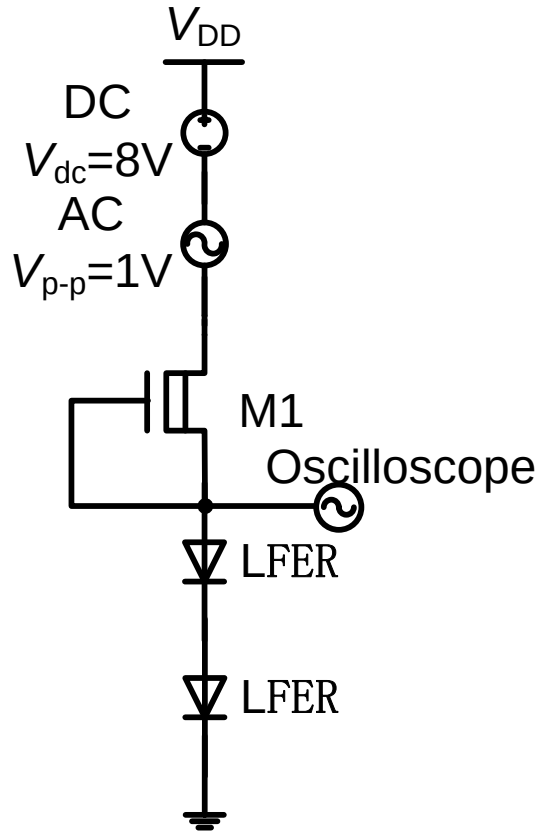


Fig. 3.4.3 Schematic of on-chip testing platform.

The measured results are illustrated in Table 3-2 and Fig. 3.4.4. An average value of PSRR is -37.98dB. Additionally, the temperature-dependent PSRR is also investigated and shown in Table 3-3 and Fig. 3.4.5. As temperature rises from RT to 250 °C, there is no obvious difference in PSRR.

Table 3-2 PSRR of GaN-based temperature sensor at room temperature.

Frequency (kHz)	10	20	30	40	50	60	70
PSRR (dB)	-38.06	-37.89	-37.89	-38.06	-37.89	-37.89	-37.89
Frequency (kHz)	80	90	100	200	300	400	500
PSRR (dB)	-37.89	-37.89	-37.89	-37.89	-37.89	-37.89	-37.89
Frequency	600	700	800	900	1000		

(kHz)							
PSRR (dB)	-37.89	-37.89	-37.89	-35.78	-36.48		

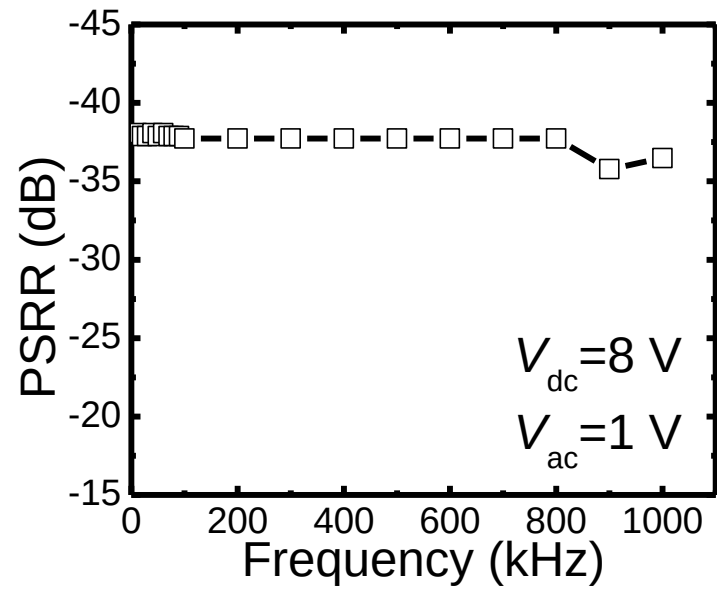


Fig. 3.4.4 PSSR of temperature sensor as a function of frequency at $V_{DD} = 8\text{ V}$

Table 3-3 Temperature-dependent PSRR of GaN-based temperature sensor

Temperature (°C)	RT	100	150	200	250
PSRR (dB)	-38.06	-38.06	-38.12	-38.10	-38.19

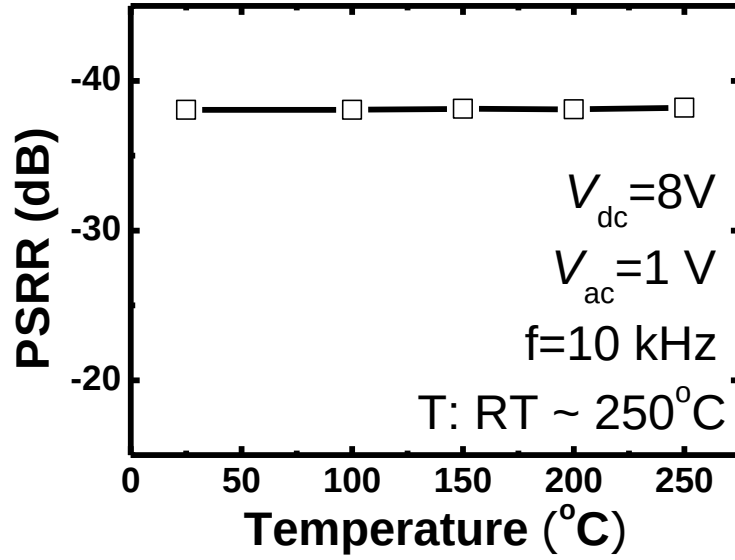


Fig. 3.4.5 Temperature-dependent PSRR of GaN-based temperature sensor

3.5 Summary

A GaN-based temperature sensor is achieved on standard GaN-on-Si AlGaIn/GaN wafers. It is presented to convert temperature changes into an output voltage, a stable wide temperature range and excellent high-temperature stability. The sensor can deliver a stable negative temperature coefficient as $-6.81 \text{ mV}/^{\circ}\text{C}$ with an error of $0.04 \text{ }^{\circ}\text{C}/\text{mV}$. The proposed sensor is HEMT-compatible, in addition, the simple structure makes monolithic integration with other GaN-based ICs possible.

Chapter 4 GaN-based PTAT Temperature Sensor

4.1 Motivation for Design of PTAT Temperature Sensor

The sensing principle of most of the present-day integrated smart temperature sensors on silicon-based platform is based on the use of PTAT signal, which depends directly on the absolute temperature and the ratio of the current densities. Additionally, it is independent of technology parameters, which has less demand on absolute accuracy in fabrication process.

A schematic of popular silicon-based basic PTAT current generator is shown in Fig. 4.1.1. It was first described by Van Kessel and Van der Plasche in 1971 [32].

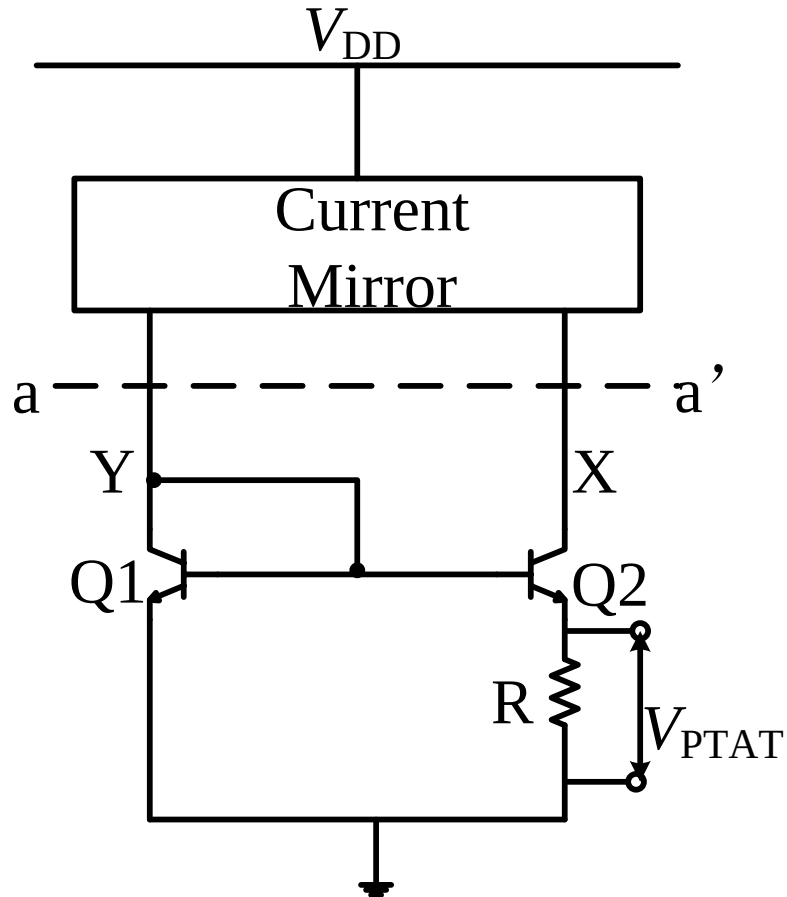


Fig. 4.1.1 Schematic of basic PTAT current generator.

The section above a-a' is a negative current mirror, which was originally realized by a pair of common base-connected PNP-type bipolar transistors to play as a current mirror [33]. It aims to ensure X and Y are at the same voltage level. As a consequence, the voltage applied to R is the same voltage as the difference of base-emitter voltages (ΔV_{be}) of Q1 and Q2, which is proportional to absolute temperature. The section below a-a' is Widlar circuit [34] in which Q1 and Q2 have an emitter area ratio of 1 : m. Ideally, the collector current I_c of a bipolar transistor in the forward active region is related to the base-emitter voltage V_{be} . This forward voltage V_{be} is used as a measure of temperature:

$$V_{be} = [(kT)/q] \ln(I_c / I_s)$$

The difference ΔV_{be} (V_{PTAT}) between Q1 and Q2 which is matched with different current densities is:

$$V_{PTAT} = [(kT)/q] \ln[(I_{c1}A_{e2}) / (I_{c2}A_{e1})]$$

It is obvious that a good linearity of V_{PTAT} to temperature relies on the degree of matched voltage between X and Y. It has been reported that using only a current mirror is not sufficient to ensure the same voltage between X and Y. Thus, a novel methodology called 4 transistors (4T) cell self-biased differential common gate amplifier [35]-[37] is proposed to obtain better matching voltage (Fig. 4.1.2). By applying this design, good matching can be promised by fabrication. In addition, it simplifies the design of the amplifier in temperature sensor and effectively reduces the power consumption, compared with other sensor designs with a current mirror driven by the operational amplifier (op amp).

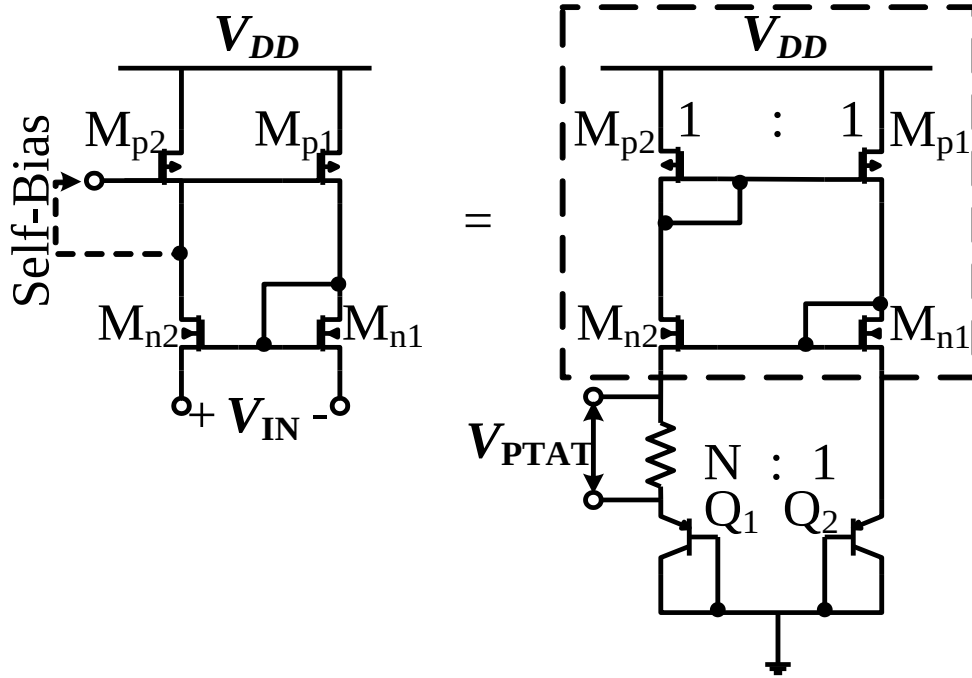


Fig. 4.1.2 4T cell self-biased differential common gate amplifier based PTAT voltage generator.

Through continuous evolution, the PTAT temperature sensor dominates the mainstream of the silicon-based temperature sensing technology by the merit of distinct operating principle, simple supporting circuits, easy to integrate monolithically, excellent temperature dependence characteristic and linearity. Also, the result is easy to read without need for complex converting interface, which, to a large extent, facilitates signal process.

However, the temperature sensor on the GaN-based technology was not previously developed on PTAT sensing principle. The main design challenge is due to lack of p-type devices.

In this chapter, a PTAT temperature sensor is proposed on the AlGaN/GaN platform by using all HEMTs and SBDs which is GaN-based devices compatible. A good linearity with a stable positive coefficient temperature sensor is presented, which can operate up to 275 °C, with a temperature coefficient of 2 mV/°C and an average error of 0.02 °C. Then, an average of the PSRR of the proposed PTAT temperature

sensor is reported as -52.77 dB. Both temperature range and coefficient are greatly improved compared to silicon-based PTAT temperature sensor.

4.2 PTAT Temperature Sensor Structure and Operating Principle

To achieve sensed temperature proportional to the absolute temperature, the bandgap temperature sensor featuring p-n junctions and transistors is the preferred configuration [38][39]. However, in GaN technology, the p-n junction diodes and p-type transistors have not been successfully integrated with the HEMT structure. In this work, we aim to develop a PTAT temperature sensor using the conventional HEMT and SBD structures. Fig. 4.2.1 depicts the PTAT temperature sensor circuit with two SBDs (D1, D2) with different active areas as the sensing part and an all D-mode current mirror, where cross-gate-connected HEMTs is designed to transfer an equivalent current from one sensing branch to another. The E-mode HEMT pair M5 and M6 force the same voltage at V_x and V_y as temperature varies.

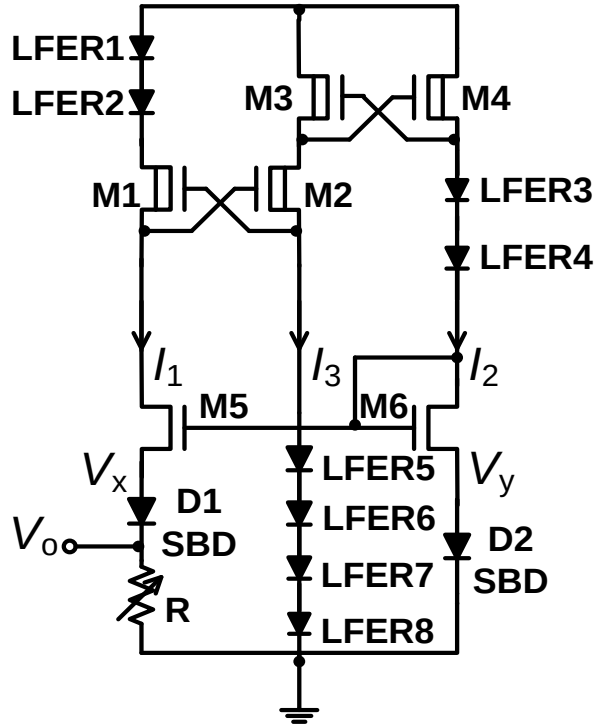


Fig. 4.2.1 Structure of PTAT temperature sensor.

4.2.1 All-D-mode Current Mirror

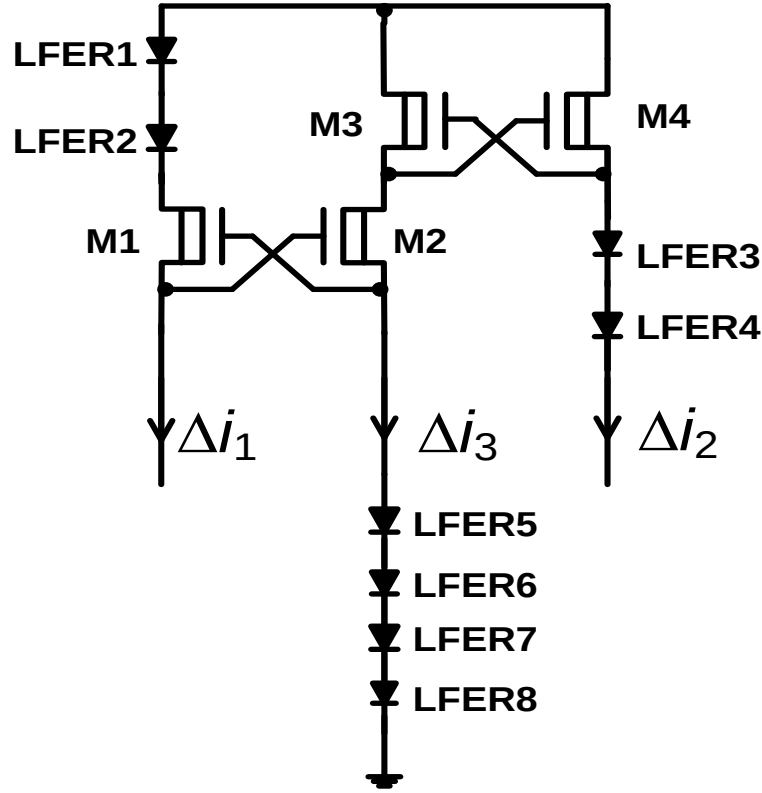


Fig. 4.2.2 Schematic of all-D-mode current mirror.

Due to the shortage of high-quality P-type transistors in the GaN technology, the all-D-mode current mirror is derived from a previous current mirror on the GaAs. Since there are also no P-type devices on the GaAs heterostructure platform, which coincide with GaN platform, Toumazou proposed the first GaAs current mirror demonstrated by D-mode MESFET in 1990 [39]. However, there are some adjustments for the single polarity supply in this application.

The core part of the current mirror was two pairs of D-mode HEMTs, whose gates and sources cross coupled (Fig. 4.2.2). To reduce the offset of fabrication, all-D-mode HEMTs are identical of 15 μm wide and 2 μm long for proper operational current. The M3 was cascaded to M2 and static current of all branches should maintain the same. Therefore the $V_{gs1} = -V_{gs2} = -V_{gs3} = V_{gs4} = 0$ V at static status ideally. When small variation occurs at load M1, its V_{gs} would be altered which induces the current I3

changes in an opposite way. Consequently the M2 transistor induces a similar variation in M3-M4 pair and cause change in the current I_2 in the inverse direction of I_3 . Therefore, the change of I_1 current will ultimately arouse the same change in the I_2 branch, if all these transistors have the same dimensions and match well. Further match design is achieved by LFER1, 2, 3, 4 to account for the V_{DS} voltage drop of HEMTs. All these LREFs are 60 μm wide and 1.5 μm long. The LFER5, 6, 7, 8 were inserted to realize better voltage matching between the central branch and the other two branches. In this way, a proper voltage at source of M3 can satisfy the downside circuit headroom.

4.2.2 Pair of SBDs PTAT Temperature Sensor

The core sensing part of PTAT temperature sensor is shown in Fig. 4.2.3, which consists of a pair of SBDs (D1, D2), a 2-DEG resistor (R) which can be adjusted by post-laser-trimming, and a pair of common gate connected E-mode (M5, M6) current mirror.

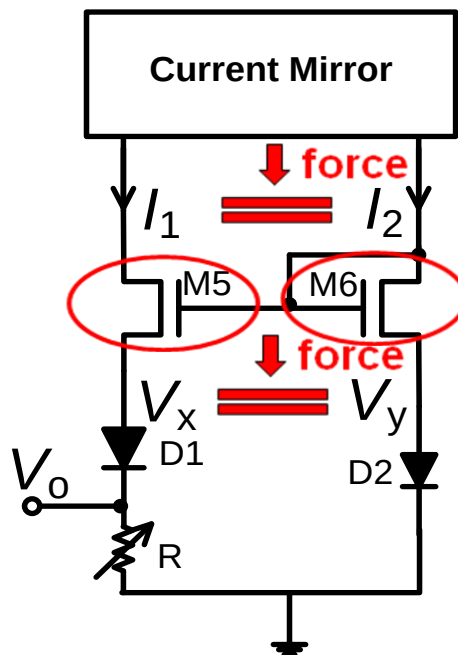


Fig. 4.2.3 Sensing part of PTAT temperature sensor

The SBD operates with a Schottky barrier which determines the temperature dependence of the current-voltage characteristic is similar to P-N junction. The voltage difference across two SBDs (ΔV_D) with different current densities is also linearly proportional to the temperature after some simplification. It can be derived from:

$$I_1 = N_1 I_{D1} = N_1 I_s \exp[(qV_{D1} - I_1 R_s)/nkT]$$

$$I_2 = N_2 I_{D2} = N_2 I_s \exp[(qV_{D2} - I_2 R_s)/nkT]$$

Where N_1 and N_2 are effective width-length ratios of D1 and D2 respectively, I_s is so called saturation current of the diode, n is the ideal factor of the diode, q is the electron charge, which equals to 1.602×10^{-19} C, and k is Boltzmann's constant, which equals to 1.38×10^{-23} J/K. By ignoring the parasitic resistance (R_s) of diode, the voltage difference of two diodes can be expressed as:

$$\Delta V_D = V_{D2} - V_{D1} = \frac{nkT}{q} \ln\left(\frac{N_1 I_2}{N_2 I_1}\right)$$

Where T is calculated in degree Kelvin, $N_{1,2}$, $I_{1,2}$, n and q are independent on temperature, so the ΔV_D is proportional to absolute temperature.

If V_x exactly equals to V_y , the voltage drops across R is obviously ΔV_D ,

$$V_o = I_1 R = V_{D2} - V_{D1} = \frac{nkT}{q} \ln\left(\frac{N_1 I_2}{N_2 I_1}\right)$$

which plays as the output voltage. Thus, M5 and M6 are inserted to force $V_x = V_y$. In order to achieve better matching, M5 and M6 are identical with $50 \mu\text{m}$ wide and $2 \mu\text{m}$ long. When $I_1 = I_2$, because of the common-gate connected, the same current can cause the same voltage between the gate and source of the pairs of E-mode HEMTs, so $V_x = V_y$ can be achieved. From another perspective, the common gate connected

pairs ensure the current in branch 2 can be mirrored to branch 1, which can promise a current mirror in a reverse way to all-D-mode current mirror.

At last, a post laser trimming technology is applied to 2-DEG resistor design in order to adjust the resistor's value after fabrication to make up the design and fabrication variation. Therefore the voltage output, PTAT, temperature sensor is realized through the voltage drops across R.

4.3 Fabrication

The PTAT temperature sensor circuits were fabricated on an $\text{Al}_{0.26}\text{Ga}_{0.74}\text{N}/\text{GaN}$ HEMT sample grown by MOCVD on a 4-inch (111) Silicon substrate. The epitaxial structure includes a 1.8 μm -thick GaN buffer, a thin AlN interface enhancement layer, 18 nm undoped AlGaN barrier and a 2 nm GaN cap layer. The starting wafer features a sheet resistance of 340 ohm/square and a pinch off voltage of -2.1 V.

The fabrication process started with the mesa isolation by Cl_2/He ion dry etching, the active regions were defined by photolithography. Then Ohmic contacts were formed using e-beam deposition of Ti/Al/Ni/Au (20 nm/150 nm/50 nm/80 nm) and subsequent rapid thermal annealing (RTA) at 850 °C for 30 seconds. The specific contact resistance was 0.85 $\Omega\text{-mm}$ obtained by using a standard transfer length method (TLM) procedure. Two separate Schottky contact were then conducted in sequence. One of them features CF_4 plasma ion implantation before the metal deposition that shifted the channel's threshold voltage to a positive value (110W for 170s) for the realization of the normally-off (E-mode) HEMTs. The other was deposited to form normally-on (D-mode) and SBDs. The self-aligned Schottky contacts were formed by e-beam evaporation of Ni/Au and liftoff. Afterwards, a silicon nitride was grown for passivation and protection purposes. Finally, a top layer of interconnection metal was deposited by e-beam evaporation after polyimide

isolation was formed [41]. In order to achieve an efficient insulation in the multi-level of the metal interconnect process, a 1.2 μm thick polyimide was inserted between the first metal layer and the top metal layer. The micrograph of a fabricated PTAT temperature sensor circuit is given in Fig. 4.3.1.

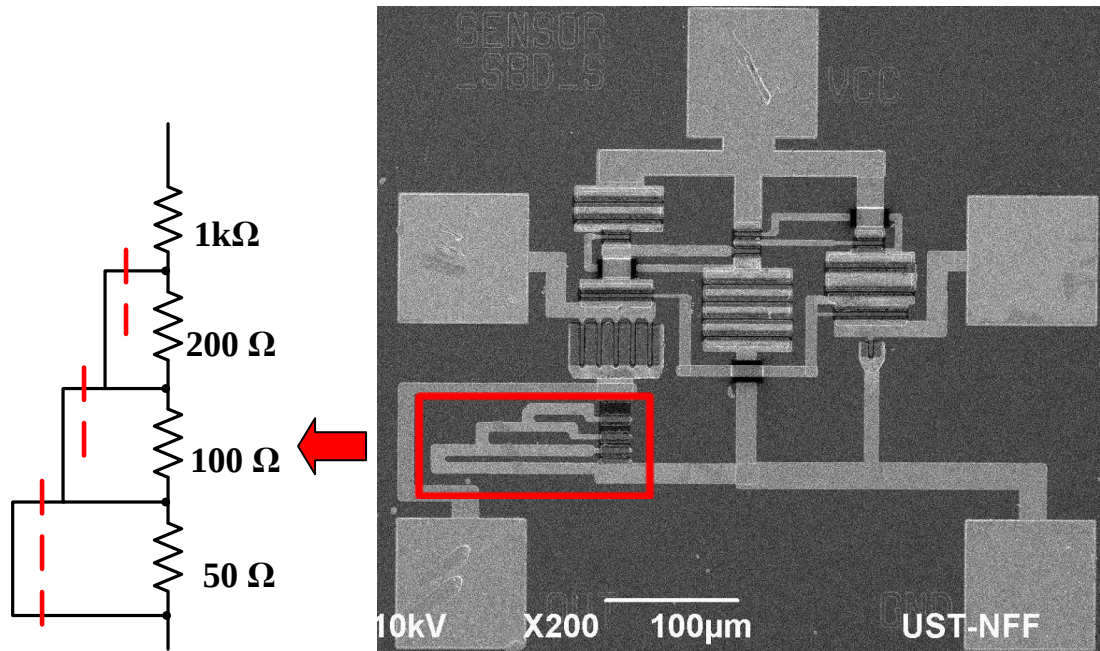


Fig. 4.3.1 SEM image of PTAT temperature sensor

4.4 Measurements and Result Analysis

4.4.1 Temperature Dependent DC Measurements

The temperature dependent $V_o - V_{DD}$ (supply voltage) characteristic of PTAT temperature sensor is measured by using a Signatone probe station equipped with Agilent 4124B Modular Source/Monitor and S-1060 Hot Chuck Temperature Controlling System. The $V_o - V_{DD}$ characteristic is plotted in Fig. 4.4.1. Due to the sophisticated cascade structure, the supply voltage (V_{DD}) is at least 13 V for temperature up to 275 $^{\circ}\text{C}$.

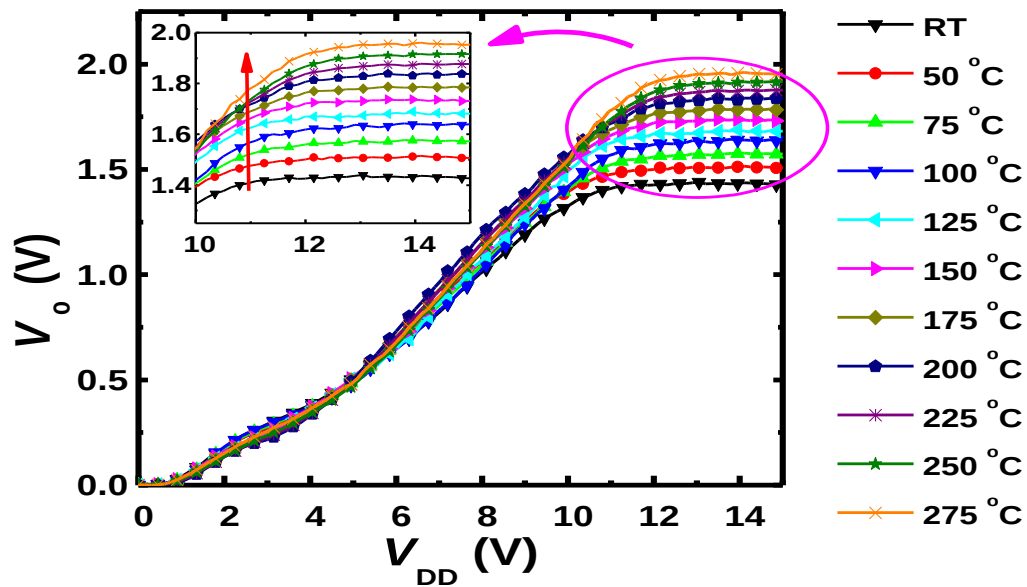


Fig. 4.4.1 Temperature dependent V_o - V_{DD} characteristic

When the temperature rises from RT (300 K) to 275 °C (548 K), the output voltage rises from 1.43 V to 1.96 V, if we set V_{DD} as 14 V. The V_o - T relationship can be extracted from $V_o - V_{DD}$ when the supply voltage is determined. The proposed temperature sensor showed a positive temperature coefficient as 2 mV/°C with a deviation of 7.18E-2 mV/°C at 14 V supply voltage. Both measurement and linear fitting results of the output voltage is shown in Fig. 4.4.2, when the temperature rises from RT (300 K) to 275 °C (548 K) with 25 °C (25 K) per step. The total voltage shift is 530 mV and an average error is 0.03 °C.

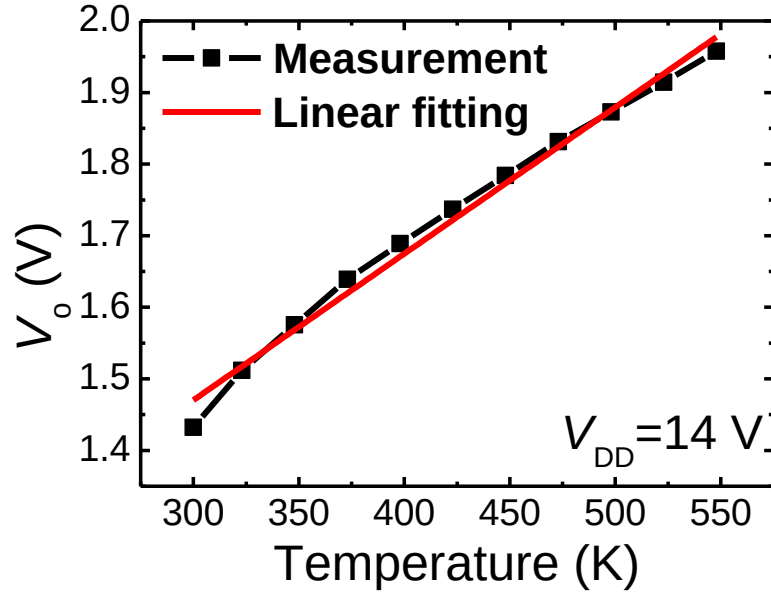


Fig. 4.4.2 V_o -T characteristic of PTAT temperature sensor

The sensing SBD is characterized in Fig. 4.4.3, which shows negative temperature coefficient between 1 and 1.5 V.

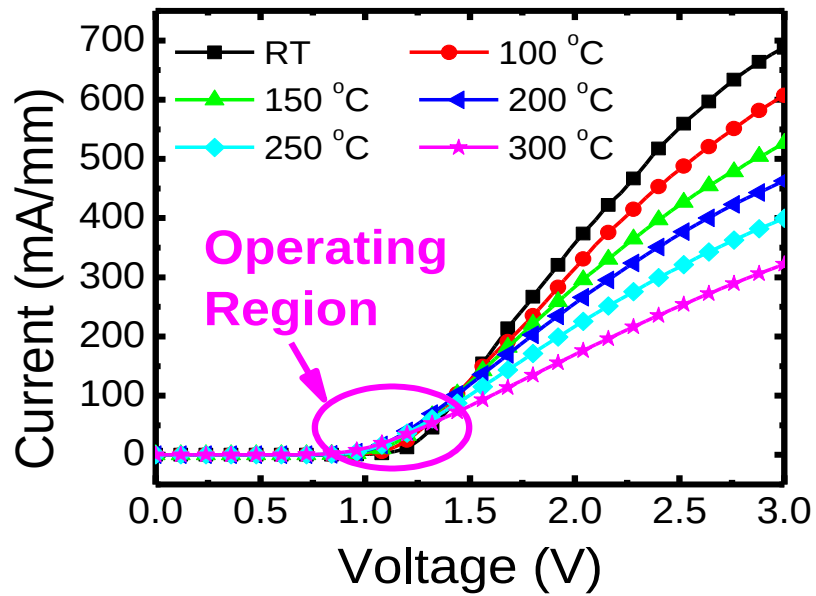


Fig 4.4.3 Temperature dependent I - V characteristic of sensing SBD

Since the main operation region lies in the sub-threshold region, SBD's current shows a positive temperature coefficient. Therefore, it can be seen that the SBD characteristic appears dominant, by ignoring contact resistance, the current flowing through SBD can be expressed as below.

$$I_1 = N_1 I_{D1} = N_1 I_s \exp(qV_{D1} / nkT)$$

$$I_2 = N_2 I_{D2} = N_2 I_s \exp(qV_{D2} / nkT)$$

$$\text{If } V_x = V_y, V_o = V_{D2} - V_{D1}$$

Unfortunately, due to fabrication offset, mismatch is incurred between V_x and V_y which is shown in Fig 4.4.4. If we express the slight difference between V_x and V_y as $\delta(T)$, which is dependent on temperature, the output voltage can be calculated as below.

$$V_x = V_y + \delta(T)$$

$$V_o = V_{D2} - V_{D1} - \delta(T) = \frac{nkT}{q} \ln\left(\frac{I_2}{N_2 I_s}\right) - \frac{nkT}{q} \ln\left(\frac{I_1}{N_1 I_s}\right) - \delta(T) = \frac{nkT}{q} \ln\left(\frac{N_1 I_2}{N_2 I_1}\right) - \delta(T)$$

$$V_o = \left(\frac{nk}{q} \ln N\right) * T - \delta(T) \text{ where } N = \frac{N_1}{N_2}$$

Hence, $\delta(T)$ is the main cause of the sensor's error.

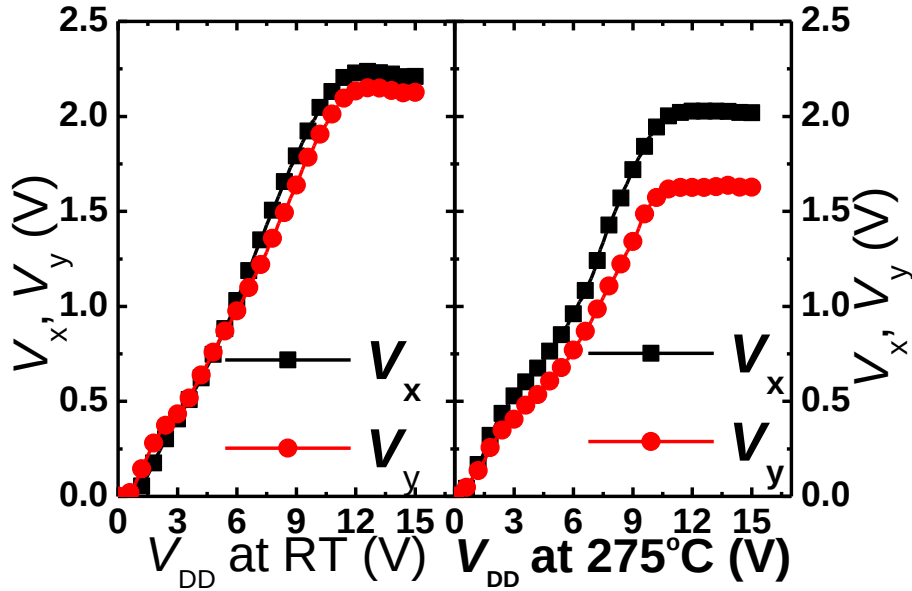


Fig. 4.4.4 $V_{x,y}$ - V_{DD} relation at RT (left) and 275 °C (right)

There are mainly three sources of this error: mismatch of current mirror, mismatch of E-mode pair, and temperature-dependent resistance of resistor. The current mirror induced error can be traced to the current difference between two branches, which is

caused by the shift of mirrored current. Since the mirrored current (I_2) shifts a 0.2 mA at a static branch current of 1.5mA compared to original current (I_1), when temperature rises from RT to 275 °C, V_x and V_y exhibit a voltage difference of 0.03 V. This variation accounts for 10% of the total voltage difference (0.3 V) between V_x and V_y . Then the 90% error is derived from the mismatch of M5, 6 and temperature-dependent resistance of resistor. The mismatch of M5, 6 can be due to the fabrication variation and slight ununiformity of the 2DEG density of the sample. And since the 2DEG resistor exhibits a large temperature coefficient which is comparable to the temperature coefficient of SBD, it can also be a significant factor to the error.

In order to improve the linearity of the PTAT temperature sensor, it is essential to reduce the mismatch between the branches of current mirror. It can be achieved through the choice of a better static bias current of all-D-mode current mirror compared to this design. Additionally, the temperature-dependent characteristic of 2DEG resistor can also cause the mismatch between two branches. It can be replaced by metal thin-film resistors with low temperature coefficient. However, due to the low resistance of metal, the area of resistor needs to be trade-off.

4.4.2 Power Supply Rejection Ratio (PSRR)

Again, to generate a reliable temperature sensor in a harsh signal environment, the temperature sensor must exhibit high dc accuracy despite supply noise. A PSSR of 52.77 dB was measured by applying an ac source to the power supply and measuring the response at the output. The measurement result is shown in Table 4-1 and Fig. 4.4.5. It can be seen that PSRR is effectively improved by cascade structure, compared to other sensor designs.

Table 4-1 PSRR of PTAT temperature sensor

Frequency (kHz)	10	20	30	40	50	60	70
PSRR (dB)	-52.77	-52.49	-52.49	-53.14	-53.14	-53.14	-52.77
Frequency (kHz)	80	90	100	200	300	400	500
PSRR (dB)	-53.14	-52.77	-53.98	-53.98	-53.98	-53.98	-52.77
Frequency (kHz)	600	700	800	900	1000		
PSRR (dB)	-52.77	-52.77	-52.77	-52.77	-52.49		

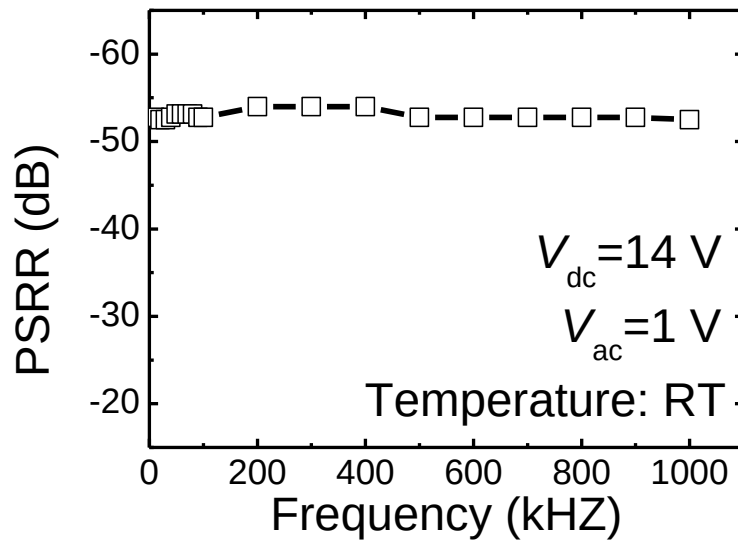


Fig. 4.4.5 Measured PSRR as $V_{DD}=14\text{ V}$ at room temperature

A temperature dependent PSRR also are measured at RT, 100 °C, 150 °C, 200 °C, 250 °C respectively, which is shown in Table 4-2 and Fig. 4.4.6. However, it influences to PSRR, because the dominant effects of PSRR are slightly temperature dependent, such as the output resistance of the device at saturation region.

Table 4-2 Temperature-dependent PSRR of PTAT temperature sensor

Temperature (°C)	RT	100	150	200	250
PSRR (dB)	-52.77	-54.19	-53.22	-53.06	-51.27

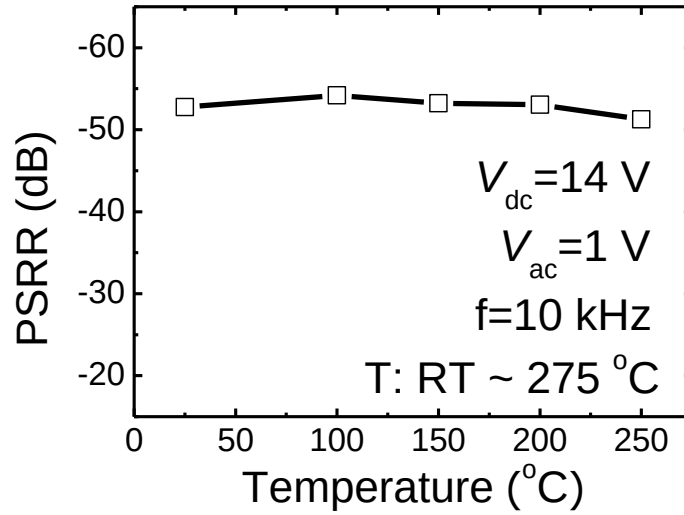


Fig. 4.4.6 Temperature dependent PSRR characteristic at 10 kHz

4.5 Summary

A PTAT temperature sensor is realized on the GaN-based platform. The circuit is fabricated on standard GaN-on-Si AlGaIn/GaN wafers. It performs a stable wide temperature range and good linearity. A stable positive temperature coefficient of 2.0 mV/ $^{\circ}\text{C}$ with an error of 0.03 $^{\circ}\text{C}/\text{mV}$ can be achieved. The main sensing part is designed with the SBD which is HEMT compatible, and a pair of matched current is provided by all-D-mode HEMT current mirror. These designs, somehow, eliminate the main obstacle to develop the GaN-based temperature sensor. Additionally with the cascade structure of current mirror, an optimized PSRR is delivered by PTAT temperature sensor. A comparison between two sensors is show in Table 4-3.

Table 4-3 Comparison of two types of sensors on GaN-based platform

Figure of merit	Type 1	Type 2 (PTAT)
Output range (V)	1.71	0.53
Temperature coefficient (mV/°C)	-6.81	2
Error (°C/mV)	0.04	0.03
PSRR (dB)	-38.06	-52.77
Minimum power consumption (mW)	4.5	54

Chapter 5 Over-temperature Protection

5.1 Motivation

GaN-based power devices are emerging as attractive candidates for next generation power converters since they can be operated at high current, high voltage and high frequency levels, especially higher temperature. A high-efficiency power conversion ratio can be achieved under certain extreme harsh environment. However extremely high temperature caused by over-current or extreme ambient temperature may lead to failure of appropriate operational function even irreversible degradation of the power device. Thus, it is desirable to develop over-temperature protection for smart power chip technology with large thermal operation region and accurate critical protection point. In this chapter, we demonstrate an analog functional block, over-temperature protection with changeable preset cut-off temperature, on a proposed GaN smart power chip technology platform. It includes a GaN-based temperature sensor which was introduced in Chapter 3 and a previous proposed single polarity power supply bootstrapped comparator [31]. An accurate protection of up to 275°C can be delivered.

5.2 Design and Working Principle of Over-temperature Protection

Over-temperature, as a significant part of the smart power technology, promises to provide an effective protection to power electronics when temperature rises over a critical value. The critical value is recognized as a cut-off point, if the temperature exceeds the value, it will have bad effects on the performance or operation of devices. However, when the temperature is below this critical point, the over-temperature protection circuit is supposed to have no influence on the power devices. Therefore a

buffer stage was inserted, on one hand to achieve the aim of having no influence on the power stage when temperature is below critical value, on the other, it is designed to provide a proper driving voltage and current to the power devices. A functional block diagram of the over-temperature protection is illustrated in Fig. 5.2.1.

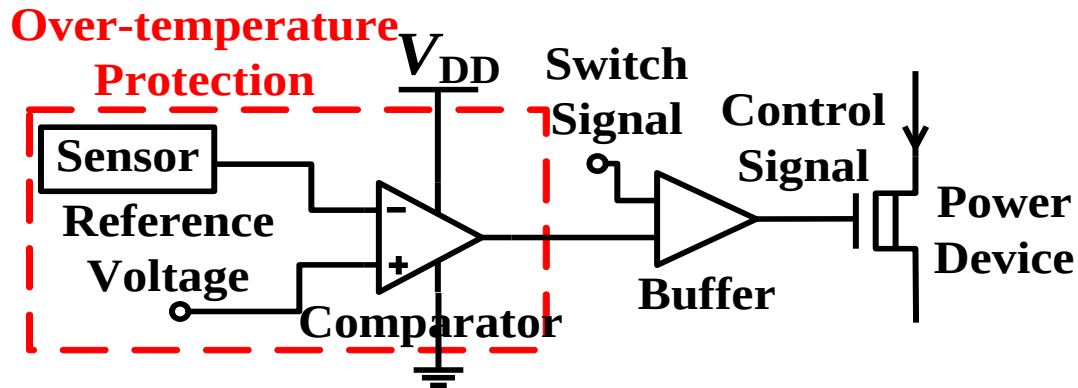


Fig. 5.2.1 Functional block diagram of over-temperature protection

The reference voltage is set as a critical voltage which is comparable to the output voltage of the temperature sensor. This critical voltage corresponds to the critical temperature value one at a time. When temperature exceeds the critical value, the output voltage of temperature sensor will exceed the reference voltage, which results in a voltage shift at the output of the comparator.

The designed structure of the over-temperature protection is illustrated in Fig. 5.2.2. It consists of the comparator (left – hand) and the temperature sensor (right – hand).

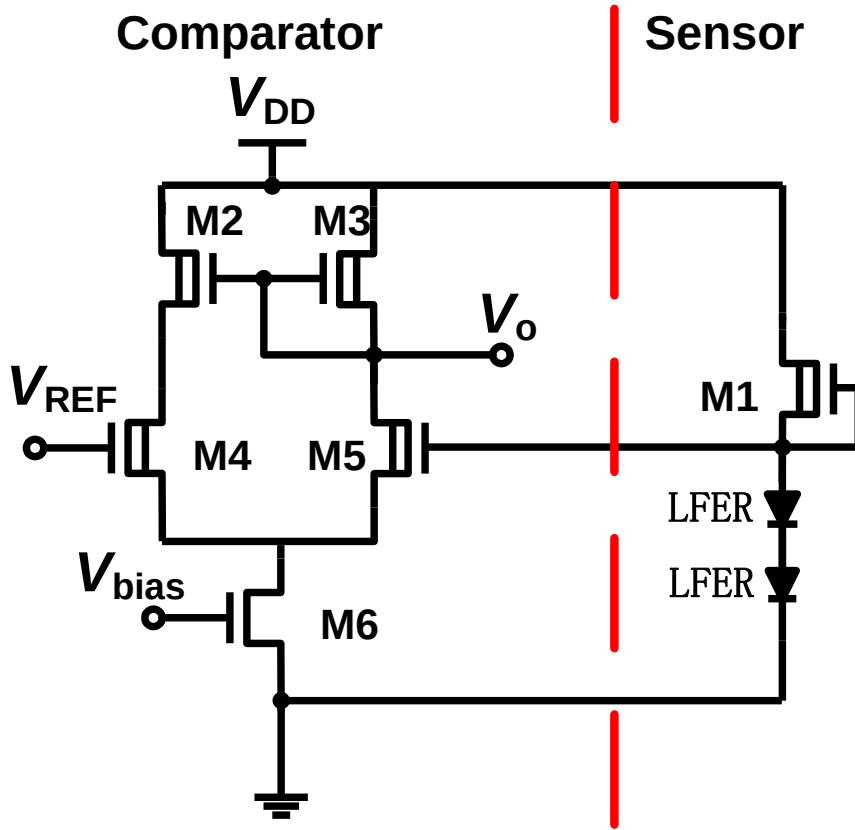


Fig. 5.2.2 Structure of over-temperature protection

The performance of the temperature sensor has been demonstrated in chapter 3. The previous proposed comparator integrates both enhancement-mode (E-mode) and depletion mode (D-mode) AlGaIn/GaN HEMTs. The active load is implemented with D-mode HEMTs using a bootstrapped structure to improve the voltage gain. The input stage is also D-mode HEMTs to cover the output range of temperature sensor, which can cover from -1 to 3 V. The tail current source uses the E-mode HEMT, which ensures the single polarity power supply. At room temperature, the comparator delivers a voltage gain as high as 132 V/V and a unity-gain bandwidth of 270 MHz. The comparator is also capable of operating at 275 °C with temperature-compensated bias.

Since the sensor's output shows a negative temperature coefficient, the output of the over-temperature protection which is also the output of the comparator can be

described as when temperature is below critical point the comparator outputs relative low voltage, when temperature rises above critical point, a relative high voltage is output from comparator. Therefore an AND logic gate will be needed as a buffer stage to provide a control signal to the power device. When the temperature is not extremely high, over-temperature protection circuit outputs a low voltage to the buffer stage and the controlling signal changes according to the switching signal, it therefore has no influence on the power device working status; however, when temperature exceeds critical point, over-temperature protection circuit outputs a high voltage to the buffer stage and the controlling signal will output a relatively low voltage to shut down power device no matter what switching signal is. In this way, an over-temperature protection is realized.

5.3 Fabrication

The sample used to fabricate the designed over-temperature protection circuit was $\text{Al}_{0.26}\text{Ga}_{0.74}\text{N}/\text{GaN}$ HEMT wafer, the same as that of the temperature sensor. The starting wafer featured a sheet resistance of 330 ohm/square, 2DEG mobility of 1800 $\text{cm}^2/\text{V}\cdot\text{s}$ and a pinch off voltage of -2.1 V.

The micrograph taken by SEM is shown in Fig. 5.3.1. The fabrication process is also the same with the temperature sensor. Firstly, mesa isolation was carried out by the inductively coupled plasma reactive ion (Cl_2/He) dry etching (ICP-RIE). Secondly, the ohmic contacts were deposited by e-beam sequence of Ti/Al/Ni/Au (20 nm/150 nm/50 nm/80 nm) and subsequent rapid thermal annealing (RTA) at 850 °C for 30 s. For E-mode HEMTs, before Schottky metal deposition, they were first treated with CF_4 plasma ion implantation at 110 W for 170 s to shift threshold voltage from -2.1 V to +0.2 V. Then, the self-aligned Schottky contacts were defined separately by photolithography to form D-mode and E-mode HEMTs. Then, the sample was

annealed at 400 °C for 10 min. to repair the plasma-induced damage in the AlGaN barrier.

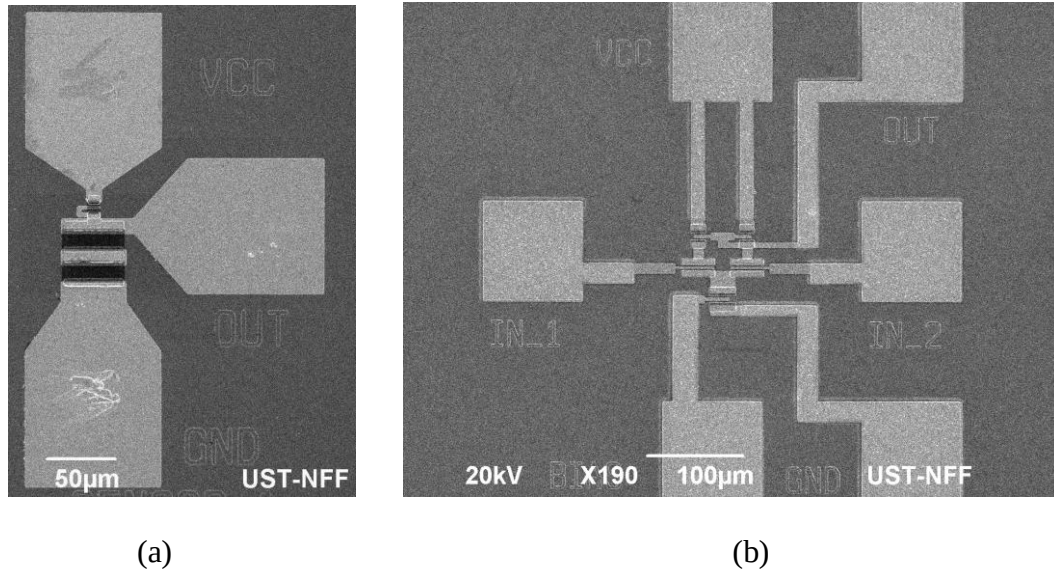


Fig. 5.3.1 SEM imaging: (a) temperature sensor; (b) comparator

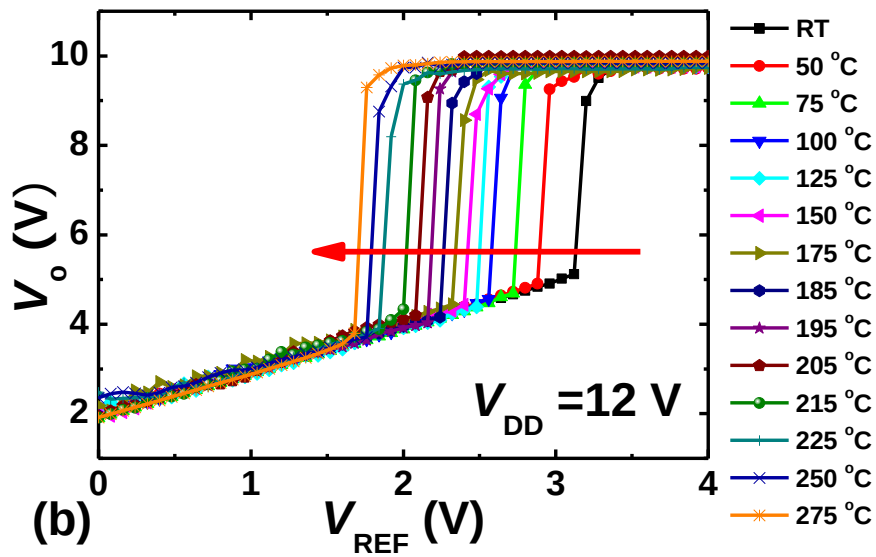
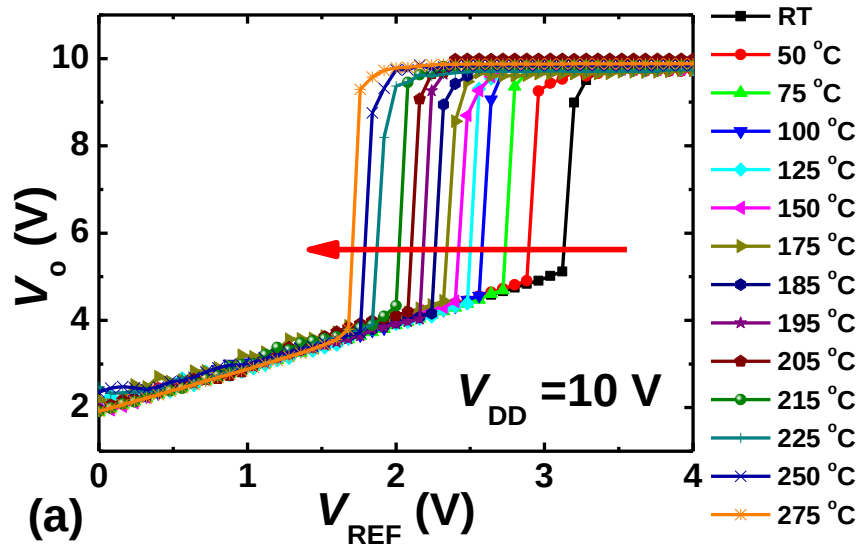
5.4 Measurement and Analysis

The temperature dependent (up to 275 °C) output voltage (V_o) is shown in Fig. 5.4.1 with 10 V and 12 V supply voltage (V_{DD}) respectively as reference voltage (V_{REF}) sweeps from 0 V to 4 V, and bias voltage is at 1.1 V.

When $V_{DD} = 10$ V and the temperature rises from RT to 275 °C, the output voltage changes as V_{REF} sweeps. If we fix $V_{REF} = 2.07$ V, when temperature exceeds 215 °C, a large swing of output voltage occurs. Details can be seen in Fig. 5.4.1 (c). The output voltage changes from 3.96 V to 9.53 V and the voltage swing is 5.6 V. The high voltage margin and low voltage margin are less than 0.5 V and 0.14 V respectively.

When $V_{DD} = 12$ V and the temperature rises from RT to 275 °C, output voltage changes as V_{REF} sweeps. If we fix $V_{REF} = 1.95$ V, it is obvious that, when temperature exceeds 215°C, a large swing of output voltage occurs. Details can be seen in Fig. 5.4.1 (d). The output voltage changes from 3.98 V to 11.74 V and the

voltage swing is 7.76 V. The high voltage margin and low voltage margin are less than 0.37 V and 0.12 V respectively.



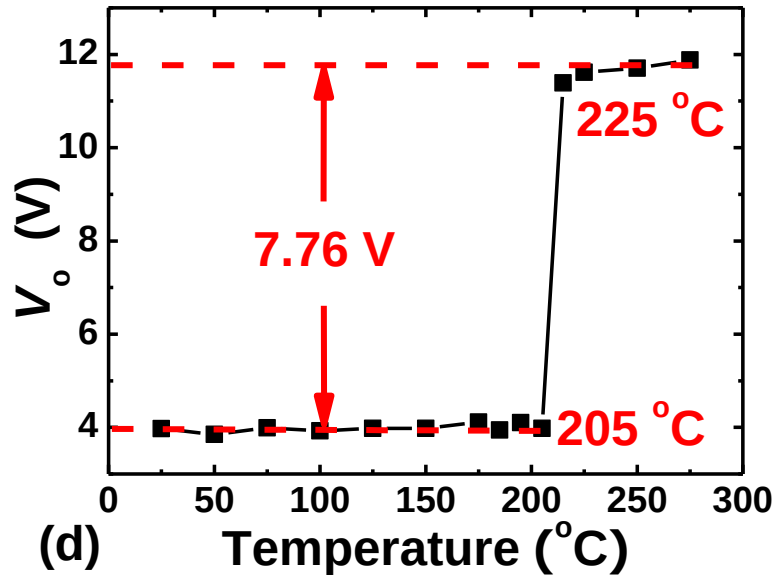
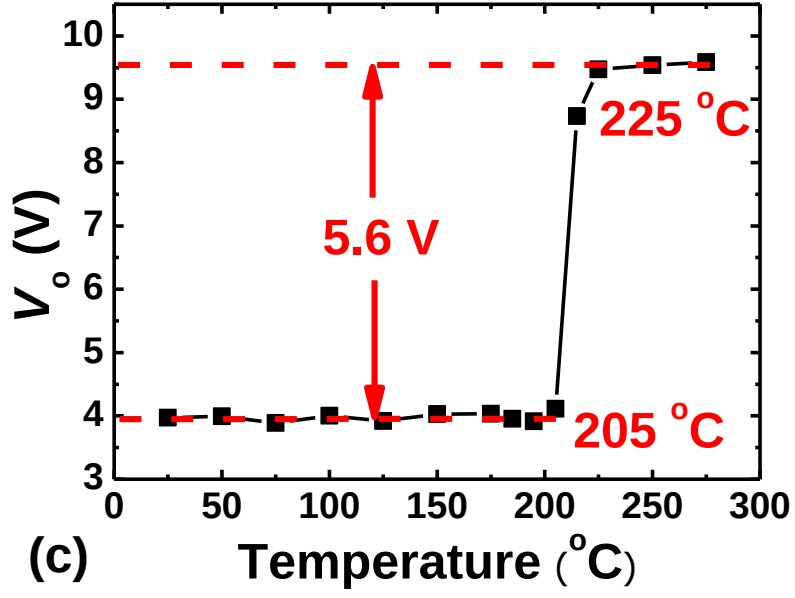


Fig. 5.4.1 Temperature dependent output voltage of over-temperature protection

(a) $V_{DD} = 10\text{ V}$, V_{REF} sweeps from 0 to 4 V; (b) $V_{DD} = 12\text{ V}$, V_{REF} sweeps from 0 to 4

V; (c) $V_{DD} = 10\text{ V}$, $V_{REF} = 2.07\text{ V}$; (d) $V_{DD} = 12\text{ V}$, $V_{REF} = 1.95\text{ V}$

Therefore, a changeable critical temperature value can be extracted through changing the reference voltage. The results are shown in Fig. 5.4.2. When $V_{dd} = 10\text{ V}$, the over-temperature protection can be realized at 185°C to 225°C with a 10 °C increase per step at each corresponding reference voltage. Similarly, when $V_{dd} = 12\text{ V}$,

an over-temperature protection can be achieved at changeable critical temperature values.

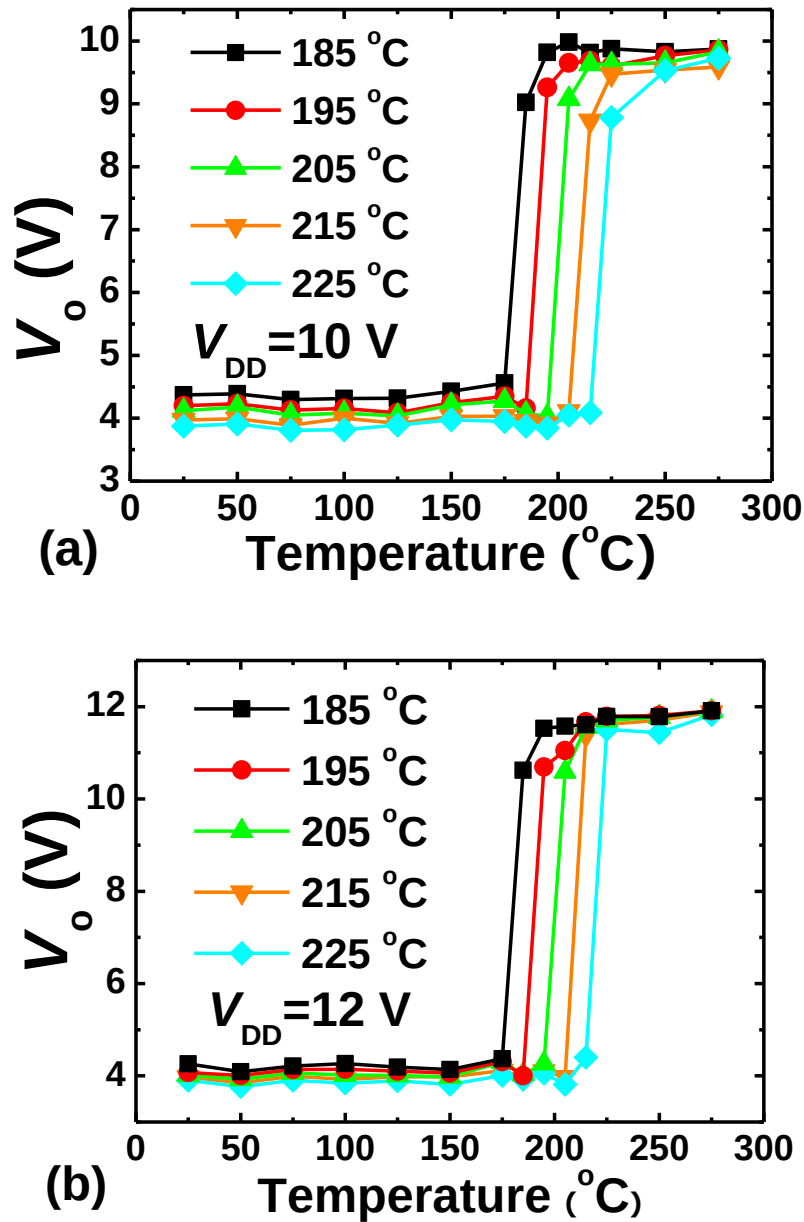


Fig. 5.4.2 Output voltage with changeable critical temperature
(a) $V_{DD} = 10\text{ V}$; (b) $V_{DD} = 12\text{ V}$

5.5 Summary

In this chapter, an over-temperature protection for the GaN-based smart power technology is presented, which consists of the single polarity power supply temperature sensor and bootstrapped comparator. It operates well up to 275 $^{\circ}\text{C}$. A

changeable protection temperature value is realized through changing the reference voltage of comparator. When the temperature rises over preset critical voltage, a voltage shift occurs through the comparator. The voltage swing is large enough to be distinguished as high or low voltage respectively to control the power device through a buffer stage.

Chapter 6 Conclusion and Suggestion for Future Work

6.1 Conclusion

As the AlGaIn/GaN heterostructure devices technology emerges promisingly and develops maturely, more focus has been placed on design of circuits and monolithic systems on the GaN-based platform. Power electronics is one of the most beneficiaries from GaN's distinct properties. However, most of these applications are under harsh environment especially in high power and high temperature applications. Therefore, an on-chip temperature sensing and over-temperature protection are indispensable for performance optimization.

In this work, three analog functional blocks are proposed on the GaN platform. They are a GaN-based temperature sensor, an over-temperature protection IC and a PTAT temperature sensor. The key objective of this thesis is to achieve a on-chip temperature sensing circuit design and fabrication without the need for p-type devices.

Initially, temperature dependent characteristics of the GaN-based devices are investigated to facilitate the design of temperature sensors. Subsequently, a GaN-based temperature sensor without complicated structures is designed and fabricated. It can output a stable negative coefficient temperature-dependent voltage and operate properly up to 275°C. Then PSRR is observed as a figure of merit to express its noisy rejection capability. The sensor has been demonstrated to work stably under noisy condition and deliver a reliable performance with different supply voltages.

Then, a PTAT sensor is realized, which converts temperature changes to an output voltage with optimized linearity. It derived from a silicon-based bandgap principle, however the executive sensing device is GaN-compatible SBD instead of bipolar transistor which is commonly used in sensor design on the silicon platform. A

matched current is implemented by a cross-coupled current mirror which applies all D-mode HEMTs and LFERs to substitute the p-type current mirror. Consequently, an output voltage which is proportional to absolute temperature with positive temperature coefficient is promised. PSRR is also improved through a cascade structure. Additionally, 2DEG resistor is used to eliminate complexity of fabrication, and post-trimming technology is also applied to changeable resistor for the purpose of better matching.

Finally, based on the design of GaN-based temperature sensor, which is introduced in Chapter 3, an over-temperature protection for smart power technology is proposed. The protection circuit consists of the proposed temperature sensor and a bootstrapped comparator. A reference voltage is provided to set a critical value for protection purpose. When the temperature is below this critical value, there will not affect the target power device. However, when the temperature rises above the critical value, the target power device will be shut down to enhance the safety and reliability. In order to adapt to different application conditions, the critical value can be preset and changed through varying the reference voltage.

6.2 Suggestions for Future Work

Although several functional blocks are proposed and discussed in this thesis, some modifications and improvements can still be developed. Additionally, more essential functional blocks need to be achieved and further integrated with the proposed ones to consummate GaN-based smart power technology and other monolithic circuit designs on this promising platform. The suggestions for future work are summarized as the following four points:

- (1) To improve the linearity of PTAT temperature sensor, an advanced current-mirror design with smaller mismatch compared to all-D-mode current mirror is needed.

- (2) A low temperature coefficient metal thin-film resistor can be applied in the PTAT temperature sensor design to further improve the linearity.
- (3) A relatively high gain and accurate amplifier needs to be realized monolithically with the PTAT temperature sensor, which can amplify the small output voltage of PTAT sensor to satisfy the further applications.
- (4) More functional blocks can be implemented to develop the smart temperature sensor, such as analog-to-digital convertor, which can facilitate temperature sensing applications.

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